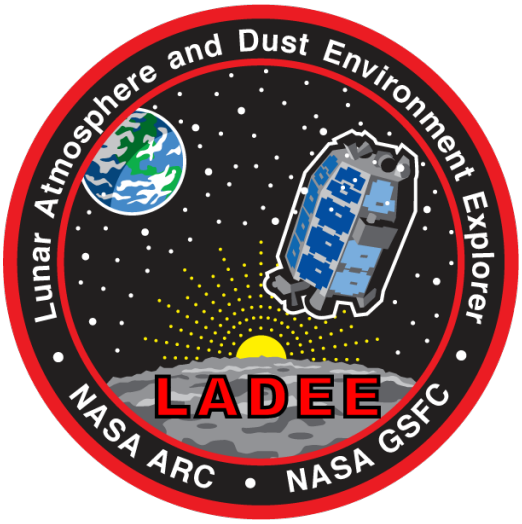


LADEE Reverse Engineering Study

Final Report

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Nomenclature

ADCS Attitude Determination and Control System

AKE Attitude Knowledge Error

AMOAB Analog Monitor and Control Board

AOCS Attitude and Orbit Control System

APAR Attitude Profile Activity Report

APE Attitude Pointing Error

ATS Absolute Time Sequence

BOL Beginning of Life

cPCI Compact Peripheral Component Interface

C&DH Command and Data Handling

COTS Commercial Off-The-Shelf

DET Direct Energy Transfer

DLAT Daylight Analysis Tool

DMOAB Digital and Mass Memory Board

DoD Depth of Discharge

DSN Deep Space Network

EEPROM Electrically Erasable Programmable Read-Only Memory

EOL End of Life

EPS Electrical Power System

FAS Frame Alignment Sequence

FDS Flight Dynamics System

FOV Field of View

FPGA Field-Programmable Gate Array

GG Gravity Gradient

GMAT General Mission Analysis Tool

IAU Integrated Avionics Unit

IMU Inertial Measurement Unit

IOC Instrument Operations Center

LASS Long-term Attitude Scheduling System

LDEX Lunar Dust Experiment

LEOP Launch and Early Orbit Phase

LLCD Lunar Laser Communications Demonstration

LLST Lunar Lasercom Space Terminal

LOI Lunar Orbit Insertion

MCSB Modular Common Spacecraft Bus

MEB Main Electronics Box

MGA Medium Gain Antenna

MIPs Million Instructions per Second

MLI Multi-Layer Insulation

MOC Mission Operations Center

MPS Mission Planning System

NMH Monomethyl hydrazine

NMS Neutral Mass Spectrometer

NTO Nitrogen Tetroxide

OBC On-Board Computer

OBDH On-Board Data Handling

OCS Orbit Control System

OMM Orbit Maintenance Maneuver

PAPI Plug-and-Play Avionics Protocol Interface

PCM Pulse Code Modulation

PID Proportional-Integral-Derivative

PLM Payload Module

PPM Pulse-Position Modulation

PPT Peak Power Tracking

PS Propulsion System

RCS Reaction Control System

ROM Read-Only Memory

RW Reaction Wheel

SA Solar Array

SACI Solar Array Charging Interface

SATORI Solid State Power Controller and Telemetry and Command Interface

SCaN Space Communications and Navigation

SCPPM Serially Concatenated Pulse Position Modulation

SDRAM Synchronous dynamic random-access memory

SN Space Network

SOC Science Operations Center

SOI Sphere of Influence

SRAM Static random-access memory

SRP Solar Radiation Pressure

SS Star Sensor

ST Star Tracker

STDN Spacecraft Tracking and Data Network

SUROM Startup Read-Only Memory

SVM Spacecraft Vehicle Module

TCS Thermal Control System

TEC Thermoelectric Coole

TCM Trajectory Correction Maneuver

TLI Trans-Lunar Injection

TMTC Telemetry, Tracking and Command

TTP Transfer Trajectory Phase

TWT Traveling Wave Tube

UVS Ultraviolet/Visible Spectrometer

VDU Valve Driver Unit

1 Mission introduction

Change log	
Section	*Comments*
1.1	Introduction is shortened
1.1.1	High-level goals are revised
1.2	Mission drivers are revised
1.3	Figure 1 related to Functional analysis is changed
1.4	System-level requirements are revised
1.5	Table 1 related to mission phases is revised with duations explicitly mentioned
1.6	Fig. 2 for ConOps is revised.

1.1 Mission description

NASA's Lunar Atmosphere and Dust Environment Explorer (LADEE) was a robotic mission that operated from September 2013 to April 2014. Its primary goal was to study the composition and dynamics of the Moon's tenuous atmosphere and dust exosphere. The mission also served as a technology demonstration, testing a novel laser communications system and a new low-cost spacecraft bus. This combination of an aggressive science campaign with new technologies created significant challenges for mission planning and execution. This study examines those challenges, the resulting operational requirements, and the key elements of the LADEE mission architecture designed to meet them.

1.1.1 High-level goals

The LADEE mission was structured around four main objectives, encompassing both pioneering scientific investigations and the validation of new technologies designed to enhance future space exploration.

- **Understand the Origins and Evolution of the Lunar Atmosphere** A key goal was to trace the history of the lunar atmosphere. This involved examining the interplay between various contributing factors, such as the solar wind, outgassing from the lunar surface and interior, and the regular influx of meteoritic material.
- **Investigate the Lunar Exospheric Dust Environment** The mission sought to gather comprehensive data on the dust present in the lunar exosphere. Insights gained from this investigation are intended to be applicable to other celestial bodies, as exospheres are the predominant atmospheric type in our solar system.
- **Determine the Composition and Dynamics of the Lunar Atmosphere** This objective focused on identifying the chemical constituents of the Moon's tenuous atmosphere. It also aimed to understand the key processes governing its structure and temporal changes, including sources, loss mechanisms, and interactions with the lunar surface.
- **Demonstrate Novel Mission-Enabling Technologies** This objective involved validating three key technological innovations. The first was the Lunar Laser Communications Demonstration (LLCD), a system for high-bandwidth optical communication. The second was a cost-effective and adaptable modular spacecraft design. The third was the confirmation of the Minotaur V rocket's capability as a launch platform for interplanetary missions.

1.2 Mission Drivers

The LADEE mission architecture was fundamentally shaped by two primary, non-negotiable drivers that dictated its trajectory, design, and operational strategy.

- **Science-Mandated Low Lunar Orbit:** The primary scientific objective required frequent measurements at low altitudes (< 50 km) over the lunar sunrise terminator, the most dynamic region for exospheric activity. This orbital requirement was a critical design constraint, as the Moon's uneven gravity field makes such an orbit inherently unstable. Consequently, the spacecraft required a robust propulsion system and a highly reactive ADCS to perform frequent orbit-maintenance maneuvers, preventing premature orbital decay.
- **Programmatic Mandate for a Modular Bus:** LADEE was required to utilize the NASA/Ames Modular Common Spacecraft Bus (MCSB), a low-cost, modular architecture. This programmatic driver was key to achieving a rapid development schedule and reduced mission cost.

However, it also imposed significant constraints on the spacecraft's overall mass, dimensions, power generation capabilities (via fixed solar panels), and payload accommodation, influencing the entire system design.

1.3 Functional analysis

System-level requirements guide the spacecraft's design, ensuring it meets performance objectives:

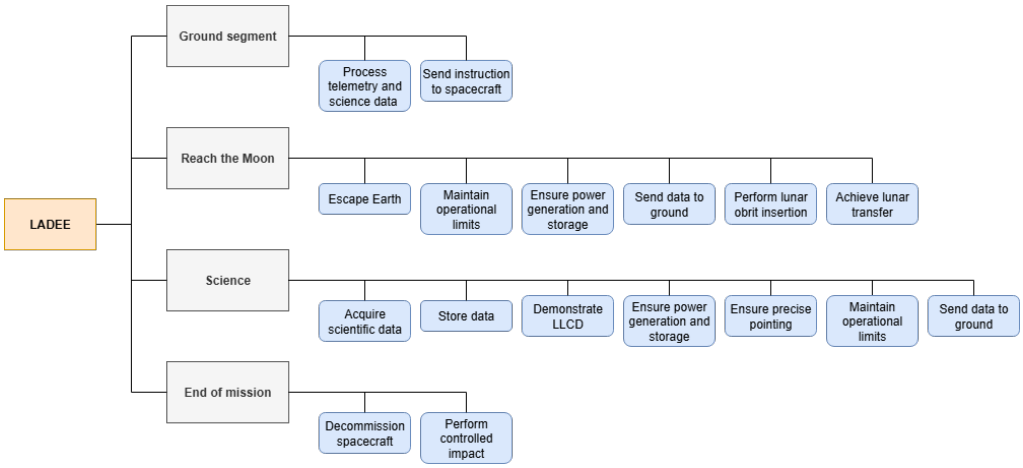


Figure 1: LADEE functional analysis diagram

1.4 System-level requirements

Requirements	Description
Functional	
A retrograde orbit shall be utilized for science data acquisition.	This is needed to better capture atmospheric particles and dust while minimizing self-contamination from spacecraft outgassing.
A goal duration of at least three months for the mission should be supported.	A longer duration allows for observing variations due to lunation, meteor showers, and solar storms.
At least one lunation of mission duration shall be provided.	It allows passing from the solar wind into the magnetotail, enabling observation of the lunar atmosphere under different plasma conditions.
Sufficient pointing precision to guarantee optimal scientific measurements shall be ensured.	-

Operational	
Efficient management of electrical power shall be performed by the spacecraft.	To ensure that mission-critical operations are not interrupted, even when solar arrays cannot provide sufficient power.
An optimal temperature range for payload utilization shall be provided by the thermal control system.	The extreme temperature variations between the dark and sunlit sides of the Moon affect the ability to perform accurate measurements.
Constraints	
A controlled impact on the lunar surface shall conclude the mission.	There is no preferred impact point, but it is necessary to consider the planetary protection guidelines.
The spacecraft's mass shall be constrained by the Minotaur V's TLI capability.	Minotaur V, for TLI, can deliver 383 kg from a circular parking orbit. This constrained the total wet mass of the spacecraft.
The spacecraft envelope shall be designed to fit within the Minotaur V payload fairing.	Minotaur V can fly with two types of fairings: the standard 92-inch (2055 mm payload diameter) and a larger 110-inch (2515 mm).

1.5 LADEE Mission Phases

The LADEE mission consisted of several key phases, each essential for the success of its objectives. The following is an explanatory summary of each phase.

Phase	Duration (approx.)	Description
Launch and Check-out Phaset	1 day	Launched on **September 7, 2013**, from Wallops Flight Facility. Following separation from the Minotaur V launcher, the spacecraft was activated in low-Earth orbit for system checkouts and initialization of ground communication
Phasing Loops Phase	24 days	Involved three phasing loop orbits around Earth to correct trajectory deviations. This phase included system tests and concluded with a maneuver toward the Moon.
Lunar Transfer and Acquisition Phase	8 days	Began on **October 1, 2013**, with injection into a trans-lunar trajectory. A series of Lunar Orbit Insertion (LOI) maneuvers captured LADEE into a circular orbit at 250 km altitude.
Commissioning Phase	40 days	Focused on system tests, instrument activation, and the LLCD technology demonstration. The orbit's periapsis was lowered to 50 km for initial science data collection.

Science Phase	100 days	Starting **November 21, 2013**, LADEE conducted detailed studies of the Moon's exosphere and dust environment, with orbital maintenance maneuvers performed as needed.
Extended Science Phase	48 days	Started on **March 1, 2014**. LADEE's orbit was allowed to decay naturally, bringing it closer to the lunar surface for high-resolution observations.
Decommissioning and Impact Phase	3–7 days	The mission concluded with a planned impact on **April 18, 2014**. Data collection continued during the final descent until impact at the eastern rim of Sundman V crater.

Table 1: Main phases of the LADEE mission.

1.6 Concept of Operations and Timeline

The following figures represent the Concept of Operations of LADEE mission.

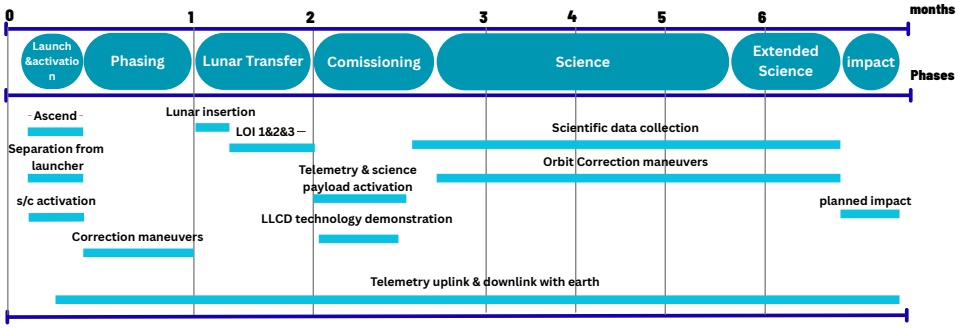


Figure 2: Timeline of mission phases. The operations carried out throughout the mission are shown based on their temporal order and divided by phases.

1.7 Payload Analysis

As mentioned previously, LADEE carried three scientific instruments and a technical one, with a total payload mass of approximately 49.7 kg. The instruments on board were: the Neutral Mass Spectrometer (NMS), the Ultraviolet/Visible Spectrometer (UVS) and the Lunar Dust Experiment (LDEX) along with the Lunar Laser Communications Demonstration (LLCD). These instruments were body-mounted on the spacecraft structure and required individual pointing, which was achieved through the spacecraft's attitude control system.

1.7.1 Neutral Mass Spectrometer (NMS)

The LADEE NMS measured the mass distribution of neutral species over a mass range between 2–150 Daltons, it used a dual ion source that provided a means to measure both surface reactive and inert species with a quadrupole mass spectrometer. NMS was operated in several modes:

- RAM mode pointed the NMS aperture in the direction of LADEE's orbital motion—this was used to track the noble gases;
- TILT mode called for LADEE to point the NMS field of view downward 30 deg off ram, which enabled NMS to measure more energetic sputtered species;

- WAKE mode was performed when LADEE pointed the aperture in the anti-ram direction, permitting measurements of instrument background when essentially no atmospheric gas entered the instrument;
- ION mode pointed the aperture toward ecliptic normal, looking for exospheric ions that have been picked up in the solar wind electric field.

1.7.2 Ultraviolet and Visible Light Spectrometer (UVS)

Its objective is to examine exospheric emissions and a broader continuum of forward or backward scattered light from dust down to 10 nm in size. The UVS has two means of observing: a Limb Telescope and a SOV (Solar Occultation Viewer). The Limb Telescope is pointed just above the surface of the Moon, at its limb, looking for emission by exosphere gasses and the scattering of sunlight by dust grains. Limb observation will be centered on the morning and evening terminators and local noon. During each of these activities the LADEE spacecraft will keep the telescope pointed no more than 20 km above the lunar surface. The SOV allows UVS to stare at the Sun, monitoring it as it rises or sets across the lunar limb.

1.7.3 Lunar Dust Experiment (LDEX)

LDEX is designed to map the spatial and temporal variability of the dust size and density distributions in the lunar environment. LDEX is an impact ionization dust detector, sensing dust impacts in situ at LADEE orbital altitudes of 50 km and below, with a particle size range of between 100 nm and 5 µm. Nominal LDEX operations were carried out when LADEE was in ram attitude, and were generally scheduled whenever it was safe to operate, that is, when the LDEX FOV did not include the sun. However, the LDEX team also wanted off-ram measurements, including anti-ram measurements to determine background levels. Because of the low LDEX power requirements and desire for of sun-safe (such that the instrument could be safely operated), off-ram measurements, it had a higher duty cycle than the other instruments.

1.7.4 Lunar Laser Communication Demonstration (LLCD)

The goal is to demonstrate and validate duplex optical communications from a lunar orbiting spacecraft to an Earth-based ground receiver. A LLCD operation consisted in 13 to 15 minutes of "on" time of the LLST with no active science instruments. Power constraints limited the number of LLCD operations that could be run in sequence before a dedicated orbit for recharging the batteries was required. During the technology demonstration, the spacecraft had to be pointed at the Earth with an accuracy error of <1 degree.

1.8 Payload Con/Ops

During the commissioning phase the LLCD technology demonstration and basic checkout of the science payload was completed. The prime operational phase of the LLCD was conducted during the first 30 days while LADEE was in the 250 km circular orbit at 157° inclination. During the science phase, both NMS and LDEX were usually operated in the preferred ram orientation, with their boresights aligned with the velocity vector of the spacecraft in orbit. However, at times both made use of off-ram activities. UVS was most often operated in a limb-stare mode, with the telescope field of view fixed on a selectable grazing point altitude above the limb. UVS also performed occultation activities with its solar viewer optic. Thus, UVS activities generally placed the spacecraft attitude several degrees off ram, generally outside the useful range for NMS measurements. This mutual exclusion of attitude requirements, considering also the energy constraints, led to instrument operations plans

that alternated UVS and NMS activities. LDEX operations were instantiated for those times when the instrument field of view did not include the sun. During real-time operations, when spacecraft engineering, data downloads and command sequence uploads took place, the spacecraft was in an attitude that pointed LADEE's medium gain antenna (MGA) at the Earth; no science activities were permitted at these times.

1.8.1 Mission Analysis Introduction

LADEE was launched on the Minotaur V, a launch vehicle composed of 5 solid propellant stages. The use of solid propellant on every stage made it impossible for the launch system to compensate for any off-nominal condition encountered during ascent. As a consequence, both the spacecraft and the lunar transfer strategy were designed to be able to compensate for off nominal launch conditions. After the first 4 stages were depleted LADEE was put into an almost circular low Earth orbit. Then the fifth stage was ignited to move the spacecraft into a highly elliptical phasing orbit where LADEE separated from the 5th stage (T+23 minutes) and performed orbit determination and ground station acquisition.

In the following two weeks the spacecraft performed an apogee burn and 2 perigee burns using the on-board bipropellant engine, to slowly raise the apogee of its orbit. This transfer strategy was selected because it provided multiple opportunities to correct errors in previous burns or deriving from the launch vehicle. Following these manoeuvres, the spacecraft entered the sphere of influence of the moon about 30 days after launch. The lunar encounter was from the leading edge and with a retrograde hyperbolic trajectory. Capture in an elliptical lunar orbit was achieved with a burn executed after passing behind the moon and re-gaining contact with the ground station. A series of manoeuvres were then performed to achieve a 250 Km circular orbit denoted as commissioning orbit. The spacecraft spent 40 days in this orbit carrying out checks and tests of various systems and scientific instruments. The commissioning orbit was chosen to provide Earth visibility during manoeuvres and minimize orbital perturbations. Finally on November 21st, 2013, the LADEE spacecraft moved to its operational orbit, where it conducted its scientific operations for 100 days. The operational orbit was a retrograde equatorial orbit characterized by a 50 km perigee, a 150 km apogee and a 2 hours orbital period. The selected orbit required many orbital correction manoeuvres because of the non-uniform gravitational field of the Moon. The main effect of the perturbation is a reduction of the perigee and an increase of the apogee. This characteristic was exploited to have the spacecraft impact the moon without using any propellant.

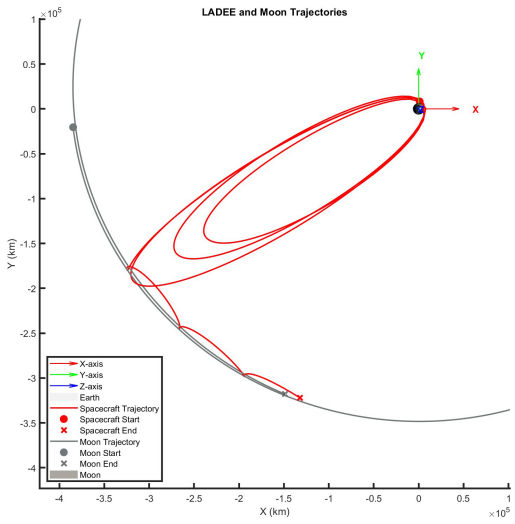


Figure 3: LADEE trajectory to reach the moon.

2 Mission Analysis

2.1 Objective of the Analysis

The mission analysis for LADEE focused on designing a trajectory that conformed to key constraints, including fuel limitations that necessitated a minimum-energy transfer, precise lunar orbit requirements, and the need to avoid prolonged periods in shadow. This analysis ensured that the spacecraft could successfully reach the Moon, conduct its science operations, and maintain orbital stability while accounting for uncertainties in launch performance and lunar gravitational variations.

2.2 Trajectory Overview

LADEE was launched on a Minotaur V rocket into an elliptical Earth orbit. Instead of a direct transfer to the Moon, the mission used a 3.5 phasing loop trajectory. This approach involved three elliptical Earth orbits with progressively raised apogees before the final lunar transfer.
The phasing loop strategy was chosen over a direct transfer primarily to accommodate potential launch dispersions from the Minotaur V rocket, allowing trajectory adjustments after launch to ensure precise lunar arrival, while also improving fuel efficiency by leveraging lunar gravitational perturbations to raise apogees instead of relying solely on propulsion.

Orbit	**Perigee Altitude** [km]	**Apogee Altitude** [km]
Initial Earth Orbit	200	275000
Phasing orbit 1	2544-2062	275000
Phasing orbit 2	2062	318,000
Phasing orbit 3	2062	375,000
Lunar Insertion	688	14869
Intermediate Moon Orbit	250	2189
Commissioning Orbit	234	294
Science Orbit	30-50	75-150

Table 2: LADEE Orbits Properties.

2.3 Orbital segments

The trajectory of LADEE was simulated using data about orbit parameters obtained from the literature. Table 2.2 summarizes the properties of each orbit.

2.4 Simulation, assumptions and approximations

The phasing manoeuvres, as well as the transfer from the commisioning orbit to the science orbit, are modeled as elliptical transfers between elliptical orbits.

For the lunar insertion maneuver, the trajectory was assumed to be a Hohmann transfer from Earth to the moon, with a hyperbolic trajectory starting from the moon sphere of influence. This simplification does not take into account the effect of Earth's gravity after the spacecraft enters the SOI of the Moon. OLM3 and OLM4 refer to the Orbit Lowering Maneuvers to move from the Commissioning phase to the Science phase by lowering the perigee and apogee, respectively. The required change in velocity due for lunar capture was obtained using the following formula:

$$\Delta\nu = \nu_{p_{\text{hyp}}} - \nu_{p_{\text{capture}}} = \sqrt{\nu_\infty^2 + \frac{2\mu_{moon}}{r_p}} - \sqrt{\frac{\mu_{moon}(1+e)}{r_p}}$$

To estimate the delta-V required for orbit maintenance of the science orbit, a simulation was set up using NASA's General Mission Analysis Tool (GMAT). The simulation was designed to propagate the spacecraft's orbit over the 100-day science phase, applying correction manoeuvres whenever the periapsis or apoapsis exceeded the ranges obtained from the analysis of LADEE's telemetry.
The simulation used the GRGM900 model (degree and order 180) to account for the Moon's gravitational field and included the perturbative effects of the Sun's and Earth's gravity.

Analysis of LADEE's telemetry data indicates that the periapsis altitude was maintained between 20 km and 50 km, while the apoapsis altitude varied between 75 km and 150 km, both are plotted with respect to time in fig. 5. Finally, to estimate the attitude control budget requirement for this mission, literature data was obtained from Lunar Reconnaissance Orbiter (LRO) mission, where the spacecraft used 30.8 kg for fuel over 14 years for attitude control solely.
The results are shown in Table 2.4. The final ΔV is higher than the actual allocated one by 20.7%. The biggest source of error between the simulated results and the real values are the standard budget margin, for example, extra 82 $\mathrm{m\,s^{-1}}$ was solely added as a margin to orbit maintenance.

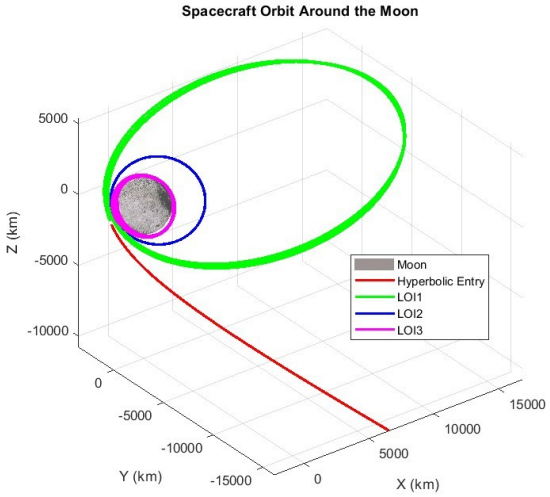
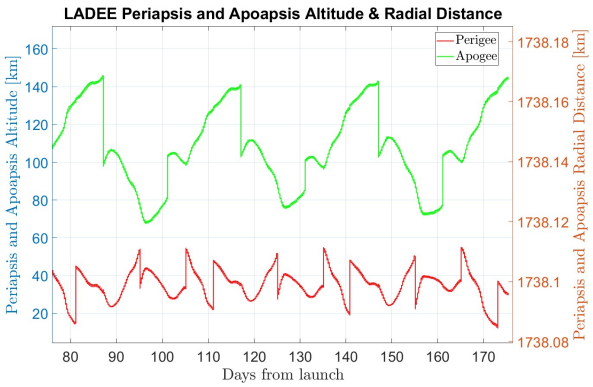


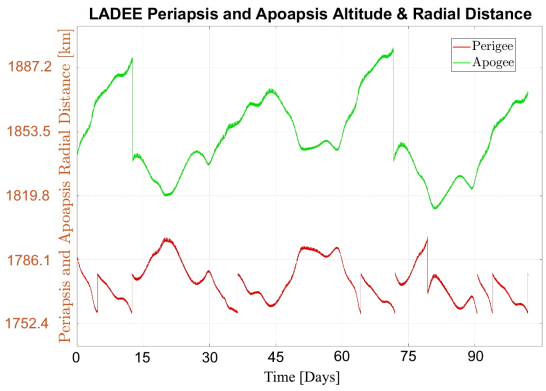
Figure 4: Moon insertions.

Maneuver	ΔV_{sim} $\mathrm{m\,s^{-1}}$	ΔV_{real} $\mathrm{m\,s^{-1}}$
Deterministic Maneuvers		
LOI 1	507.24	333
LOI 2	292.2	292.9
LOI 3	240.9	243.36
AM1B	6.935	9.35
PM1	16.762	16.97
PM2	17.443	17.49
OLM3 + OLM4	63.12	69.20
Total Deterministic	**1144.6**	**982.27**
Total with margin	**1201.8**	**1028**
Stochastic Maneuvers		
Attitude Control	4	-
Orbit Maintenance	82	101.37
Total Stochastic	**86**	**101.37**
Total with margin	**172**	**110**
Δ V Total	**1373.8**	**1138**

Table 3: LADEE List of Maneuvers.



(a) Real science phase



(b) Simulated science phase

Figure 5: Science phase comparison.

3 Propulsion system

LADEE features a bipropellant propulsion system designed for both high and low Δv maneuvers. It comprises a main thruster and four smaller attitude control thrusters. Pressurized with helium gas, the system can operate in both blowdown and pressure-regulated configurations. The system is depicted in Fig. 6.

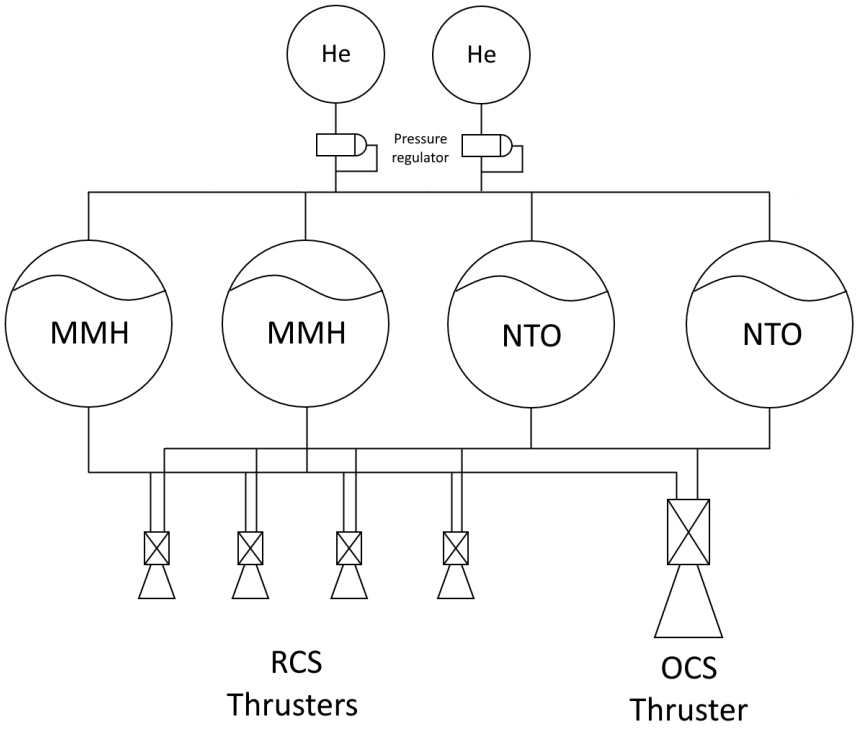


Figure 6: LADEE propulsion system diagram.

3.1 Propulsion architecture

The system consists of two fuel tanks, two oxidizer tanks, two pressurization tanks, one primary thruster, four secondary thrusters, and feeding lines equipped with valves and sensors. The propellant and pressurant tanks are arranged in a square configuration on the baseplate to ensure rotational symmetry. This layout was driven by the need to integrate the tanks within the existing spacecraft structure (see Fig. 7) while maintaining the center of mass at the spacecraft's midpoint.

3.2 System components

Due to limited information available in the literature regarding the system's components, the following description presents the known specifications alongside reasonable assumptions based on typical propulsion hardware.

- Propellant tanks: the system includes four spherical titanium tanks mounted on the base plate, with two designated for fuel and two for oxidizer. To ensure structural integrity, all tanks have the same volume and are symmetrically arranged around the center;
- Pressurization tanks: two smaller spherical titanium tanks store helium, chosen for its low molar mass, which enhances pressurization efficiency. However, its use comes with drawbacks, including

high cost and a tendency to leak;

- Pressure regulation: the system was initially operated in a pressure-regulated configuration but was switched to a blowdown mode before the first orbit maintenance maneuver;

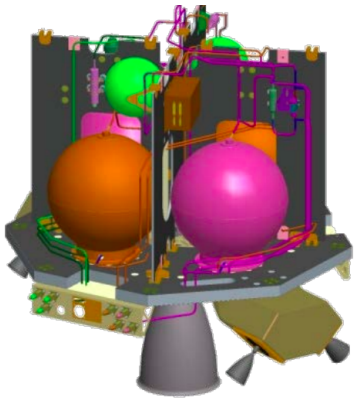


Figure 7: LADEE propulsion module.

- Primary propulsion: a single R-4D-15 HiPAT engine, constructed from platinum-rhenium, pointed along the vehicle -Z axis,delivers 455 N of thrust with a specific impulse of 323 s. This higher thrust capability is utilized for high Δv maneuvers. Developed by Aerojet Rocketdyne, it is commonly employed in spacecraft for orbit insertion and trajectory adjustments;
- Secondary propulsion: the spacecraft is equipped with 4 Aerojet Rocketdyne R-6D.The thrusters provide a thrust of 22 N with a specific impulse of 294 s. These RCS thrusters are distributed in two canted pairs (45 degrees from the Z axis) beneath the spacecraft's lower deck in a way to enable control along three axes. They are used to desaturate the reaction wheels, to perform orbital maintenance maneuvers ($\sim 10\,\mathrm{m\,s^{-1}}$) and finally to act as ullage motors.

3.3 Oxidizer and fuel selection

The choice of monomethyl hydrazine ($CH_3(NH)NH_2$) as the fuel and dinitrogen tetroxide (N_2O_4) as the oxidizer stems from the need for a propulsion system that is quick to develop, cost-effective, and reliable. This propellant combination is widely used in propulsion systems, ensuring both reliability and low cost. Additionally, as a hypergolic propellant, it eliminates the need for ignition, allowing for multiple engine firings.

3.4 Propulsion subsystem reverse sizing

As described previously, the propulsion system counts four tanks for storing propellant and oxidizer and two tanks for storing helium. To better understand the reasoning behind this choice, a reverse sizing for both the propellants and the pressurizer has been conducted given the data on the engines, the ΔV budget of the mission and the total dry mass M_{dry} of the spacecraft. The whole process has taken into account proper margins. All the data used for the following calculations were obtained from datasheets or detailed technical descriptions of the individual components, with the exception of the main engine —a high-performance 455 N HiPAT (High Performance Apogee Thruster). Since this engine was a custom modification based on the standard 445 N HiPAT, no specific datasheets were available. Therefore, data from the base 445 N version were used as a reference.

3.4.1 Propellant selection and masses

To estimate the propellant masses, the Tsiolkovsky rocket equation was applied separately to both the primary and secondary propulsion systems to determine the mass ratio between the wet and dry mass. The evaluation was based on the actual dry mass at launch, taken as $m_{dry} = 240.7\,\text{kg}$, due to the absence of reliable information on the nominal spacecraft mass. Consequently, no additional margins were applied to this value, as it was assumed to already include the necessary system-level margins.

$$\text{MR}_{OCS} = e^{\frac{\Delta V}{g_0 I_s}\Big|_{OCS}} \qquad \text{MR}_{RCS} = e^{\frac{\Delta V}{g_0 I_s}\Big|_{RCS}} \tag{1}$$

$$m_{wet_i} = \text{MR}_i \cdot m_{dry} \tag{2}$$

The total mass of the propellants, as well as the individual masses of the fuel and oxidizer, can now be determined by specifying the mixture ratio so that the volumes are equal using the propellant densities. It is important to note that a margin must be applied to the propellant mass to account for residual propellant, loading uncertainties, and ullage conditions. An important consideration must be made regarding the Excess Propellant Capacity requirement: the propellant tank volume shall accommodate an additional 10% of propellant beyond what is required to meet the ΔV requirement, assuming a proportionate increase in the allowable propellant mass.

$$m_{prop_i} = m_{wet_i} - m_{dry} \tag{3}$$

$$m_{MMH} = \frac{m_{prop}}{1 + O/F} \qquad m_{NTO} = m_{prop} - m_{MMH} \tag{4}$$

The estimated masses of fuel and oxidizer can now be computed, obtaining the following values, along with a comparison with the real masses used in the mission:

	Estimated mass [kg]	Real mass [kg]	Relative error [%]
m_{MMH}	51.9	51.2	1.3
m_{NTO}	84.9	82.1	3.4

Table 4: Real and estimated masses of oxidizer and fuel

Once the total mass is known, calculating the total volume is straightforward, after which a margin is added. With this information, we can begin the tank sizing process, taking into account the relevant design considerations and the necessary margin (MAR-CP-010):

Margin	$V_{NTOreal}$	$V_{MMHreal}$
10%	$0.0649\,\text{m}^3$	$0.0649\,\text{m}^3$

Table 5: Volume of oxidizer and fuel

From the previously determined volume of fuel and oxidizer, assumed to be equal to the tank volume, it is possible to compute the tanks radii:

$$r = \left[\frac{3(V \cdot 0.5)}{4\pi}\right]^{\frac{1}{3}} \tag{5}$$

By consulting data-sheets and technical documentation, it was confirmed that the propellant tanks where spherical and made of titanium. Given its high strength-to-weight ratio, corrosion resistance,

and proven applicability in the space industry, the Ti-6Al-4V alloy was selected as the final material. This choice ensures structural reliability while meeting the demanding operational requirements of the mission. In the following table the relevant characteristics of the selected titanium alloy are presented:

	σ [MPa]	ρ [kg m^{-3}]
Ti-6Al-4V	950	4430

Table 6: Ti6Al4V characteristics.

The required thickness for a spherical tank can be determined based on the maximum admissible stress of the selected titanium alloy and the Maximum Expected Operating Pressure (MEOP), which is 250 psi. However, to ensure reliability and mitigate eventual damages from malfunctions in the transmission lines or pressurization control valves, the MEOP is increased by a 50% safety factor. As a result, the final sizing pressure considered for the tank is 375 psi. Proceeding with the sizing, the thickness of the tank can be obtained as:

$$t = \frac{P_{in} \cdot r}{2\sigma} \tag{6}$$

It is now possible to determine the effective mass of the tanks, knowing that the propellant tanks will be of the same size and of the same material.

$$M_{tank} = \rho_{Ti} \cdot \frac{4\pi}{3} \cdot ((r+t)^3 - r^3) \tag{7}$$

The obtained results can be tabulated as follows:

	Radius [m]	Thickness [mm]	Mass [kg]
NTO	0.1979	0.2692	0.5875
MMH	0.1979	0.2692	0.5875

Table 7: Propellant tank sizing

3.4.2 Pressurant selection and masses

The propulsion system initially operates with regulated propellant pressure before transitioning to a blowdown mode prior to the first Orbital Maintenance Maneuver, with a pre-maneuver fill fraction of 23%. As part of the workflow, both the pressure-regulated and blowdown systems will be sized separately. The larger values obtained from each approach will then be selected to ensure the system can sustain both operating modes. As a first approximation, helium is treated as an ideal gas, and the tank temperature T_{tank} is assumed to be 293 K. To determine the pressure at the moment of the switch, we assume that the pressurant, at that instant, is in equilibrium with the propellant. Consequently, the initial pressure for the blowdown pressurization is calculated based on the operational pressure. Initially set at 250 psi, the pressure then gradually decreases to 160 psi. Based on these assumptions, we begin sizing the pressure-regulated system, determining the total required helium mass and volume in the process. The pressure in the pressurant tank starts at 3500 psi and gradually decreases to 345 psi over the course of the mission.

$$M_{gas} = \frac{P_{tank} \cdot (1 - 0.23) V_{prop\,real}}{R_{He} \cdot T_{tank}} \cdot \frac{\gamma}{\left(1 - \frac{P_{gas\,f}}{P_{gas\,i}}\right)} \tag{8}$$

$$V_{gas} = \frac{M_{gas\,real} \cdot R_{He} \cdot T_{tank}}{P_{gas\,i}} \tag{9}$$

After completing the sizing of the pressure-regulated pressurization system, we proceed with the blowdown system sizing. The results are then compared to determine the final design.

$$V_{prop\,BD} = 0.23 \cdot V_{prop\,real} \qquad V_{gas\,i} = \frac{V_{prop\,BD}}{\frac{P_{gas\,i}}{P_{gas\,f}} - 1} \qquad M_{gas\,BD} = \frac{V_{gas\,i} P_{tank\,i}}{R_{He} T_{tank}} \tag{10}$$

By comparing the values obtained from the sizing of both the pressure-regulated and blowdown systems, as shown in the following table, it is evident that the pressure-regulated system imposes higher requirements on mass and volume. Consequently, the final sizing will be based on this configuration.

	M_{gas} [kg]	Margin	$M_{gas\,real}$ [kg]	V_{gas} [m^3]
Regulated	0.4046	20%	0.4855	0.0122
Blowdown	0.1366	20%	0.1639	-

Table 8: Pressurant characteristics.

Through equations 5, 6 and 7 used before for the propellant, we can evaluate radius, thickness, and mass of the pressurant tank.

Radius [m]	Thickness [mm]	Mass [kg]
0.1199	1.5223	1.2329

Table 9: Pressurant tank characteristics - pressure regulated

3.4.3 Total mass of the propulsion system

The total mass of the propulsion system is calculated by considering the masses of the primary and secondary thrusters, the propellant tanks, and the pressurization system (including both gas and tanks), excluding the masses of the pipes and valves due to the lack of reliable information. A margin is then applied to account for uncertainties.

M_{tr1}	M_{tr2}	$M_{tank\,He}$	$M_{tank\,NTO}$	$M_{tank\,MMT}$	$M_{gas\,real}$	Margin	M_{PSreal}
5.44 kg	$4 \cdot 0.454$ kg	$2 \cdot 1.2329$ kg	$2 \cdot 0.5875$ kg	$2 \cdot 0.5875$ kg	0.5719 kg	10%	13.9082 kg

Table 10: Propulsion system masses.

3.4.4 Power Requirement for the propulsion system

The power requirement of the propulsion system was sized on the basis of the requirements of the thrusters and the pressurization system. Given the absence of detailed information, basing on the fact that the system is pressure-regulated and must be activated and deactivated once, it likely includes impulsive pyrotechnic valves, mechanical pressure regulators, and latch valves. Considering also past missions using pressure-regulated pressurizing systems, we estimated a possible set of values.

Primary thruster	Secondary thruster	Feeding system	Total power budget
46 W	5 W	$\sim$ 30 W	$\sim$ 96 W

Table 11: Power requirements.

4 TMTC subsystem

Change log	
Section	*Comments*
4.1	Changed paragraph wording
4.7.1	Changed wording about signal modulation
4.7.2	Changed paragraph wording, changed ground station antenna diameter and parameters linked to it, added reference for omni-directional antenna gain formula
4.7.3	Modified table 14
4.7.4	Added table 15

4.1 TMTC subsystem

The LADEE Telemetry, Tracking, and Command (TMTC) subsystem was responsible for communicating the spacecraft's status, transmitting scientific data, and receiving commands. Communications were conducted in the S-band frequency for both uplink and downlink. the TMTC design prioritized reliability through the selection of heritage S-Band technology and redundant antennas. The selection of S-Band technology was driven by its proven maturity and compatibility with NASA's existing Near Earth Network (NEN) and Deep Space Network (DSN) infrastructure.

The final TMTC Subsystem design for LADEE mainly consisted of:

1. **S-Band STDN-Compatible Transponder**: This link is compatible with NASA's already existing ground station.
2. **Antennas Configuration**: One (MGA) and two (LGAs), MGA is used for transmitting only and is placed in the upper part of the s/c while the two LGAs are divided between the top and bottom part of LADEE, providing coverage from all orientations. MGA is mainly used for high data rates evident in science data, while the LGAs are for general telemetry that is needed always.

The subsystem supported command uplink at 2 kbps and telemetry downlink at selectable data rates between 2 kbps and 128 kbps. The system also enabled two-way coherent Doppler tracking, tone tracking, and sequential ranging. The two omnidirectional low-gain antennas, positioned respectively on the bottom and on the radiation panel on top of LADEE, supported receive and, when commanded, transmit operations. Their configuration ensured continuous communication coverage, which was critical during launch and lunar orbit insertion, and enabled reliable command reception regardless of the spacecraft's orientation. The medium-gain directional antenna, also mounted on the radiation panel, which had to be directed towards Earth to archive high data rates, was efficient for downloading scientific data during the science phase. The integrated transponder design reduced integration complexity and risks, aligning with the mission's rapid development schedule. Furthermore, the design addressed the varying communication needs across different mission phases, from low-rate command and telemetry during critical maneuvers to higher-rate data downlink during science operations. Finally, The constant orbital motion of LADEE around the Moon created a challenging communication scenario with continuously changing relative positions between the spacecraft, the Moon, and Earth ground stations. The rationale for using DLAT was its ability to model these complex, time-varying geometries by integrating trajectory data with communication system parameters. From a systems engineering perspective, this was valuable because it allowed engineers to precisely analyze how the communication link geometry looks throughout the mission.

4.2 LLCD

The Lunar Laser Communication Demonstration (LLCD), developed by NASA in collaboration with MIT Lincoln Laboratory, was a secondary payload on the LADEE spacecraft designed to showcase high-speed optical communication over lunar distances. Its core component, the Lunar Lasercom Space Terminal (LLST), supported downlink rates up to 622 Mbps and uplink rates up to 20 Mbps in the 1550–1568 nm band, using pulse-position modulation (PPM), advanced error correction (SCPPM), and two-way time-of-flight ranging via frame alignment sequence (FAS) correlation. LLST operated four days per week during both the commissioning and post-primary mission phases. While LLCD enabled a parallel, high-capacity link between the spacecraft and the Mission Operations Center, the ground segment did not depend on it, as it was strictly considered a technology demonstration rather than an operational system.

4.3 TMTC Usage by Phase

During the whole mission the TMTC system handles the communication dedicating to each phase a particular configuration of the different elements present on board. The satellite utilizes all the antennas throughout the different phases of the mission. During each one, the TMTC system must satisfy different linking requirements. Considering the elements previously discussed, in the following table is depicted a distinction on which elements are used in each phase, addressing the choice with proper justifications.

Mission Phase	TMTC Components	Antenna(s) Used	Purpose and Justification
Launch & Ascent	OBC, S-band transponder, TM encoder	LGA Omni	Wide beam coverage ensures RF link regardless of attitude; supports basic telemetry.
Earth–Moon Transit	OBC, TM encoder, TC decoder, S-band	LGA Omni (primary)	Maintains comms during dynamic attitude changes and trajectory correction burns.
Lunar Orbit Insertion (LOI)	OBC, S-band, TMTC encoder/decoder	LGA Omni (fail-safe)	Ensures robust contact during orbit insertion burn; fallback if pointing lost.
Commissioning	OBC, TM encoder	LGA Omni (initial), MGA Nadir (later)	Starts with omni for stability, switches to MGA for higher bandwidth and pointing precision.
Science Operations	OBC, LLCD (science TM), S-band (housekeeping)	MGA Nadir (primary), LGA Omni (backup)	Stable pointing allows MGA use; LGA retained for safe-mode fallback.
Safe Mode	OBC, basic TM encoder	LGA Omni	Maintains Earth link regardless of spacecraft attitude; supports recovery and fault handling.
Deorbit / Impact	OBC, final command handling, TM encoder	LGA Omni	Ensures telemetry continuity even in uncontrolled reentry phase.

Additional remarks can be done about the configuration choices, in fact it can be seen how the dual omnidirectional LGA offered robust contact regardless of spacecraft orientation (essential during dynamic or failure) prone phases (launch, safe mode, deorbit), also the MGA nadir antenna provided better link margin and higher data rates during science ops when pointing could be controlled. The use of omnidirectional antennas during dynamic or uncertain attitude conditions, combined with high-gain directional links during stable phases, provided the necessary robustness and performance.

4.4 Data volume

LADEE's communication system utilized both coherent and non-coherent modes through its transponder, enabling two-way Doppler measurements essential for orbit determination. Downlink data rates reached at least 2 kbps when using the omnidirectional antennas, and ranged from 50 to 128 kbps with the Medium Gain Antenna (MGA). Uplink rates were configurable at either 1 or 2 kbps. Science data was primarily downlinked nightly during two scheduled Deep Space Network (DSN) contact windows, each lasting approximately one hour between 5:00 PM and 12:00 AM Pacific Time. These sessions were staffed and made use of the spacecraft's MGA to transmit data at the maximum rate of 128

kbps. Additional unattended "Lights Out" contacts with DSN and the Near Earth Network (NEN) were conducted throughout the day to support spacecraft health monitoring and collect tracking data. These communications relied on the omnidirectional antennas and were managed through automated ground software, which could notify operators in the event of anomalies.

4.5 Ground Segment

The LADEE Ground Segment was composed of several NASA centers and support facilities. The Mission Operations Center (MOC) managed real-time operations, mission planning, spacecraft engineering, and flight dynamics. The Science Operations Center (SOC) at NASA Goddard Space Flight Center (GSFC) handled strategic science planning, coordinated the Science Operations Team and Principal Investigators, and was responsible for distributing instrument data to the individual Instrument Operations Centers (IOCs) located at each instrument's home institution.

4.6 LADEE Communications Networks

LADEE made use of all three NASA SCaN networks in addition to the commercial Universal Space Network (USN) for tracking services:

- Deep Space Network (DSN): The primary network throughout the mission, using 34-meter antennas to provide full telemetry, tracking, and commanding (TMTC) services, including two-way Doppler and sequential ranging.
- Near Earth Network (NEN): NEN provided ground-based tracking support, including Doppler measurements, range data, and auto-track angle capabilities. It was mainly used during early phases of the mission and when DSN coverage was unavailable.
- Space Network (SN): SN provided additional command and telemetry services through the Tracking and Data Relay Satellite System (TDRSS). The Space Network (SN) is a satellite-based communication system consisting of TDRS spacecraft in geosynchronous orbit. For LADEE, SN operated in legacy S-band single access (SSA) Forward-Return (F-R) mode to support mission operations during critical periods.
- Universal Space Network (now SSC): USN ground stations, operated by a private provider, were used to supplement coverage during times when DSN assets were unavailable. USN supported tracking functions such as auto-track angle measurements and two-way Doppler but did not provide command or telemetry services for the mission.

The LADEE mission was supported through all these networks, operating in different phases of the mission based on their specific capabilities and the evolving operational needs. During launch and early orbit, the Near Earth Network (NEN) and the Universal Space Network (USN) provided tracking and support. As the spacecraft transitioned to lunar cruise, the Deep Space Network (DSN) became the primary link, offering long-range telemetry, tracking, and command. Operating in a low lunar orbit, LADEE experienced frequent periods of Earth occultation, during which direct line-of-sight communication with ground stations was not possible. The use of the Space Network (SN), which relayed data via geosynchronous satellites (TDRSS), although it could not always provide coverage during these occultations, was still valuable in extending contact opportunities. Throughout the mission, USN served as a supplemental tracking resource when NASA assets were unavailable, ensuring robust coverage and continuity across all mission phases.

Antenna	Dish Size (m)	Measurement Types	Service Provider	Services Provided
AGO Santiago, Chile	9	Auto-track angles, 2-Way Doppler, Ranging	NEN	Tracking
WS1 White Sands, NM	18	Auto-track angles, 2-way Doppler, Ranging	NEN	Tracking
HBK/ZAHB Hartebeesthoek, South Africa	10	Auto-track angles, 2-way Doppler	USN	Tracking
AUWA01 Dongara, Western Australia	13	Auto-track angles, 2-way Doppler	USN	Tracking
Beam Wave Guide (BWG) Subnet **DSS-24, DSS-34, DSS-54** Goldstone, CA (GLD) Canberra, Australia (CAN) Madrid, Spain (MAD)	34	2-way Doppler, Sequential Ranging	DSN	TT&C
High Efficiency (HEF) Antennas **DSS-15, DSS-45, DSS-65** Goldstone, CA Canberra, Australia Madrid, Spain	34	2-way Doppler (all), Sequential Ranging (DSS-45, DSS-65)	DSN	TT&C
High Speed Beam Wave Guide (HSB) **DSS-27** Goldstone, CA	34	2-way Doppler, Sequential Ranging	DSN	TT&C

Figure 8: LADEE Operational Ground Stations.

4.7 System sizing

In the following we are sizing the TMTC subsystem based on data obtained from literature and assumptions made by us when such data could not be retrieved. Our aim is to verify the feasibility of uplink and downlink communication with LADEE. The sizing will consider uplink through the omni-directional antennas only, while downlink through both omni-directional and medium-gain; it will not take into account LLCD, as it was only a technology demonstration rather than a core part of the subsystem.

4.7.1 Signal properties

As stated LADEE used S-band for communication, so we assumed that the frequency used was 2.4 GHz, with a data rate of 128 kbps for the medium-gain antenna and 2 kbps for the low-gain ones in both downlink and uplink. The encoding used is unknown, so we considered convolutional encoding, since it is a common choice for space mission and fast and reliable hardware encoder and decoder exist. As stated in [69] the system supports PCM, PSK and PM modulation, but since little information on it could be obtained a conservative factor of and $\alpha_{mod} = 1$ was assumed; from this information we can compute the gross datarate R_{gross} as:

$$R_{gross} = R\frac{\alpha_{enc}}{\alpha_{mod}} \tag{11}$$

Using $\alpha_{enc} = 2$ yields gross data rates of 256 kbps for the medium/gain and 4 kbps for the low/gain. We can now compute the bandwidth B needed to transmit as:

$$B = (1 + \alpha)R_{gross} \tag{12}$$

Where α is the roll off factor, which we set to 0.5; this yields bandwidths of 393 and 6.14 kHz.

By imposing a bit error rate of 10^{-7} in downlik and 10^{-5} in uplink and considering convolutional encoding the corresponding minimum E_b/N_0 ratios are 5 dB and 4.5 dB, to which we add a margin of 3 dB, and from which we can use the following formula to compute the minimum signal to noise ratio, which are respectively 6.24 dB and 5.74 dB

$$SNR_{min} = \left(\frac{E_b}{N_0}\right)_{min} + \left(\frac{R_{gross}}{B}\right)_{dB} \tag{13}$$

4.7.2 Antennas

Even though LADEE could make use of NEN and USN smaller antennas, it relied primarily on DSN's 34 m parabolic antennas, so the sizing will take the latter into account, while the space segment used both a directional medium-gain and two omni-directional antennas, the size and type of which are unknown. Since no information on the antennas was found we could assume their properties based only on images available. From this the medium-gain one was taken to be parabolic with a diameter of 20 cm, and the other was estimated to be 1 wavelength dipole antennas, thus with a length of 12 cm. Assuming a radiation efficiency η of 0.55 for all antennas we compute the gains for the parabolic antennas as:

$$G = 10 \log \left(\eta \frac{\pi^2 D^2}{\lambda^2} \right) \tag{14}$$

and for the omni-directional[18] as:

$$G = 10 \log \left(\frac{4\pi A_{eff}}{\lambda^2} \right) \qquad A_{eff} = 0.19\lambda^2 \tag{15}$$

The resulting gains are: 56.04 dB for the ground, 11.44 dB for the medium-gain and 3.78 dB for the ommi-directional.

We can also compute the half-power beamwidth as:

$$\theta = 70 \frac{\lambda}{D} \tag{16}$$

which yields, 0.26° for the ground antenna and 47.7° for the directional antenna mounted on LADEE; such angle for the omni-directional one doesn't exist.

The ground antenna can transmit up to 20 kW but we arbitrarily considered only 100 W to be enough; it can be seen in the final result that such value is more than enough. Given the high power output, we thought that it probably used a Traveling Wave Tube Amplifier (TWTA), which has an efficiency of 60%, leading to a total power consumption of 167 W. The medium gain antenna onboard LADEE can output 5 W of power, which combined with the need for a lightweight amplifier leads us to believe a solid-state amplifier was used. This leads to a low efficiency of 20% and a power consumption of 25 W. Table 13 lists the results obtained so far.

	R_{gross}	B	$(E_b/N_0)_{min}$	SNR_{min}	G_{tx}	G_{rx}	$P_{required}$
Downlink (MGA)	256 kbps	393 kHz	8 dB	6.24 dB	11.44 dB	56.04 dB	25 W
Downlink (LGA)	4 kbps	6.14 kHz	8 dB	6.24 dB	3.78 dB	56.04 dB	25 W
Uplink	4 kbps	6.14 kHz	7.5 dB	5.74 dB	56.04 dB	3.78 dB	167 W

Table 13: TMTC sizing parameters

4.7.3 Losses

The first type of signal loss considered are cable losses which lie between 1 and 3 dB, so we chose 2 dB. Then we compute the most significant source of signal loss: free space losses, using the average distance between the Moon and the Earth of 384 400 km inside the formula:

$$L_{space} = 10 \log \left(\frac{\lambda^2}{16\pi^2 r^2} \right) = 20 \log \left(\frac{\lambda}{4\pi r} \right) \tag{17}$$

for a total of -211.75 dB.

The pointing losses have been computed only for the medium-gain downlink, as the high pointing accuracy of the ground antenna paired with LADEE's two omni-directional antennas make them negligible. With a pointing accuracy e_{point} of 0.25° given by LADEE's ADCS system [17], and a receiver beamwidth θ_{rx} of 0.26° we compute a loss of -11.09 dB with the following formula:

$$L_{point} = -12 \left(\frac{e_{point}}{\theta_{rx}} \right)^2 \tag{18}$$

Finally, we compute the atmospheric losses using the model described in [58], considering the highly exaggerated case with spacecraft elevation of 45° we obtain -1.55dB signal loss. Table 14 lists all of the losses taken into account.

	L_{cable}	L_{space}	L_{point}	L_{atm}
Downlink (MGA)	-2 dB	-211.75 dB	-11.09 dB	-1.55 dB
Downlink (LGA)	-2 dB	-211.75 dB	0 dB	-1.55 dB
Uplink	-2 dB	-211.75 dB	0 dB	-1.55 dB

Table 14: Signal losses

4.7.4 Link budget

Before computing the link budget we need to compute the noise energy N_0; to do so we use the following formula where k is Boltzmann's constant and T is the noise temperature, which we set to 21 K for the downlink and 290 K for the uplink.

$$N_0 = kT \tag{19}$$

The following table contains the values obtained so far that are used in the link budget:

	Downlink (MGA)	**Downlink (LGA)**	**Uplink**
Transmitted power (P)	5 W	5 W	100 W
TX antenna gain (G_{tx})	11.44 dB	3.78 dB	56.04 dB
RX antenna gain (G_{rx})	56.04 dB	56.04 dB	3.78 dB
Cable losses (L_{cable})	−2 dB	−2 dB	−2 dB
Pointing losses (L_{point})	−11.09 dB	0 dB	0 dB
Atmosphere losses (L_{atm})	−1.55 dB	−1.55 dB	−1.55 dB
Free space losses (L_{space})	−211.75 dB	−211.75 dB	−211.75 dB
Bandwidth (B)	393 kHz	6.14 kHz	6.14 kHz
Net datarate (R_{net})	128 kbps	2 kbps	2 kbps
Gross datarate (R_{gross})	256 kbps	4 kbps	4 kbps
RX system noise temperature (T)	21 K	21 K	290 K

Table 15: Summary of link budget

Finally we can compute the link budget for both downlink and uplink to verify that the connection can be established. We compute the actual E_b/N_0 and Signal to Noise Ratio with these formulas:

$$\left(\frac{E_0}{N_0} \right)_{actual} = \left(\frac{P}{R_{gross} N_0} \right)_{dB} + G_{tx} + G_{rx} + L_{cable} + L_{space} + L_{point} + L_{atm} \tag{20}$$

$$SNR_{actual} = \left(\frac{P}{BN_0}\right)_{dB} + G_{tx} + G_{rx} + L_{cable} + L_{space} + L_{point} + L_{atm} \tag{21}$$

The results are reported in table 16.

	actual E_b/N_0	minimum E_b/N_0	actual SNR	minimum SNR
Downlink (MGA)	9.21 dB	8 dB	7.51 dB	6.24 dB
Downlink (LGA)	30.77 dB	8 dB	29.01 dB	6.24 dB
Uplink	27.4 dB	7.5 dB	25.64 dB	5.74 dB

Table 16: Link budget results

It can be seen that all communication modes are feasible, as all the ratios are above the minimum required, although the downlink through the medium-gain antenna, due to the high data rate, gets quite close to the margin; in fact if the elevation goes below about 18° the atmospheric losses increase to the point where the link can no longer be established. This is not a problem though, as it would rule out only the few ground stations at very high latitudes close to the poles such as NASA's Near Earth Network stations in Antartica. The dowlink through the omni-directional antennas, despite the lower gain, stays well above the threshold because of the low data rate required. The uplink has no problems as the transmitted power can easily be increased since we considered only a fraction of the full power output.

5 Attitude and Orbit Control System

The Attitude and Orbit Control System (AOCS) on the LADEE spacecraft was developed to provide three-axis stabilized control throughout the mission, enabling precise pointing for scientific observations, stable communication passes, and execution of orbit maneuvers. Designed at NASA Ames Research Center, the AOCS leveraged commercially available off-the-shelf components and mission-specific software to meet the unique demands of lunar orbit operations. LADEE's AOCS architecture is based on the modular configuration of the bus, it integrates avionics, batteries, and attitude sensors into the Radiator Assembly. The reaction wheels, arranged in a pyramidal configuration to provide full three-axis torque authority with redundancy, are mounted on the Payload Module structure. This distribution ensures effective separation between sensing and actuation elements while supporting mechanical balance and control performance.

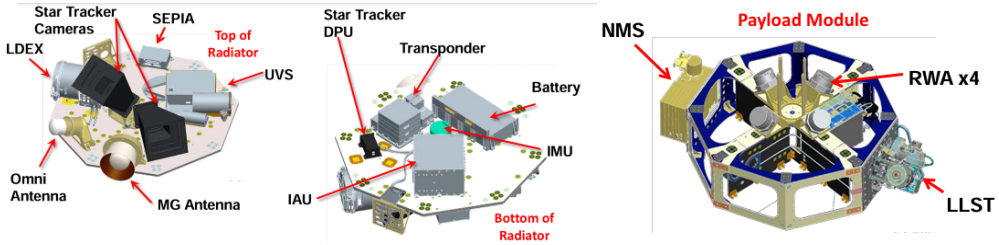


Figure 9: Internal Spacecraft Components.

Attitude planning for LADEE was particularly challenging and demanded the generation of hundreds of quaternion commands per planning cycle. These were needed to support multiple attitude configurations during the mission, each requiring different pointing geometries and constraints based on instrument needs and spacecraft orientation relative to the Sun, Earth, and Moon. Each of LADEE's instruments had distinct and often conflicting pointing requirements, this variety imposed significant challenges on planning and control strategy design. Because none of the instruments had gimbaled apertures, all pointing was accomplished via spacecraft body maneuvers, placing a high demand on the AOCS and planning tools.

5.1 AOCS Hardware: Sensors

- Dual-head star tracker: At the heart of LADEE's attitude determination system was a single dual-head star tracker (STA), which served as the primary inertial reference. The star tracker operated by identifying known constellations in the star field and comparing them to an onboard catalog, enabling highly accurate attitude knowledge. The star tracker units were mounted on radiators to manage thermal loads. The presence of two heads, with different body-frame pointing, increase the flexibility of the sensor with relation to the attitude (if one FOV is saturated by the sun the other is not).

- Inertial Measurement Unit (IMU): provided angular velocity information that allowed the onboard flight software to propagate attitude estimates between star tracker updates, this addition contributed to improve the precision of attitude determination.

- Coarse sun sensors: The Sun sensors are evenly distributed around the surface of the spacecraft and are used to define a power positive attitude relative to the Sun while the spacecraft is in Safe Mode. Coarse sun sensors guarantee low-complexity, robust sun vector acquisition for safe mode recovery and safe energy-positive pointing under all contingency conditions.

5.2 AOCS Hardware: Actuators

- Reaction wheels: The reaction wheels were used for fine attitude control during nominal operations, especially in support of science observations that required stable pointing or precise slews. The choice fell on reaction wheel rather than momentum wheels because, given the large number of possible attitude configuration, the stabilization around one axis was not a needed functionality. A CMG could also provide precise attitude control, but, as they are suitable for fast maneuvers and are more massive, again reaction wheels were preferred, because they are better suited to the low angular rates required by LADEE's star tracker during the science phase

- Thrusters: They were used to execute Orbit Maintenance Maneuvers throughout the science phase, to unload accumulated momentum from the reaction wheels, and during the initial post-separation "rate reduction" mode, where they reduced the spacecraft's body rates to within the capture range of the reaction wheels.

5.3 AOCS software

Once onboard, LADEE's Attitude Control System software used uploaded quaternion timelines to guide the spacecraft's orientation in real-time. It interpolated between time-tagged quaternions to create a continuous attitude reference, which was compared against the estimated attitude from the star tracker and IMU. A closed-loop PID controller computed torque commands to the reaction wheels to minimize the attitude error. In contingency situations, like safe mode entry, the spacecraft switched to using Coarse Sun Sensors for Sun-pointing and, if the angular rate are too high it employed the thrusters, instead of reaction wheels, until the spacecraft is slowed down enough to use the reaction wheel. Because of the large number of maneuvers required, partial automation was necessary, and a set of FDS scripts were developed to generate attitude plans from an input APAR file, which was based on scheduled activities, with prescribed attitude profiles tied to orbital events and station views, created by the LASS planning tool. Activities were mapped to the latest orbital predictions, de-conflicted, and organized into a timeline of desired attitude profiles. The FDS scripts used "aligned and constrained" attitude software to generate quaternion commands at regular intervals, meeting LADEE's pointing requirements. These quaternions were compiled into an Attitude Plan, which the MPS team then used to build the Absolute Time Sequence (ATS). Specific ground rules had to be respected, such as restrictions on when certain attitudes could be used, required slew times between attitudes, and the need for all maneuvers to start and end in the baseline Ram attitude to satisfy power and thermal needs. With these automation tools, operators could create a three-day attitude plan in just a few minutes. However, because the full planning, sequencing, and verification process still took several hours, discovering constraint violations could cause significant delays. To mitigate this, long-term attitude predictions were produced and checked early for potential conflicts.

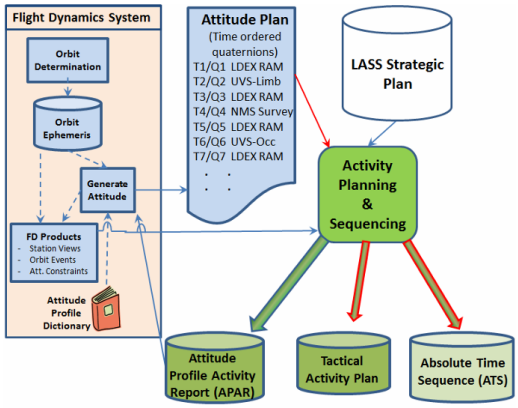


Figure 10: Internal Spacecraft Components.

5.4 Mass, power and data budgets

Literature regarding Sun sensors and star tracker used in the mission is not available so equivalent components were selected. In particular we considered the HYDRA-TC dual head star tracker from

Sodern and the cosine Coarse Sun Sensor from REDWIRE. The reaction wheels selected are the MSCI MicroWheel 1000, each equipped with a single-axis rate gyro. The thrusters used are the R-6D engines from Rocketdyne.
The data type considered for all components but the Sun sensors is float, while INT8 is used for the sun sensors since their error is $\pm 2°$.

Component	N	Mass [kg]	Power [W]	Data Size [bit]	Sample rate [Hz]	Data Rate [kbit s^{-1}]
Sun sensor	12	0.066	0	$6 \cdot 16$	10	0.96
Star tracker	2	3.9	7	$4 \cdot 32$	10	1.28
Single axis gyro	4	0.240	2	$4 \cdot 32$	100	12.8
Reaction Wheel	4	5.76	36	$4 \cdot 32$	10	1.28
Thruster	4	0.65	5	$4 \cdot 32$	10	1.28

Table 17: Mass, Power and Data Budget

5.5 Operating modes

The different pointing requirements of the scientific payload required the spacecraft to switch between multiple attitudes: to achieve this the AOCS could be operated in different modes:

- **Rate reduction mode**: this mode uses the RCS thrusters to help in attitude control. This mode was used after decoupling from the launcher to reduce the angular velocity down to a point where the reaction wheels could take over; it was also used to de-saturate the reaction wheels and to keep the spacecraft stable during main engine burns.

- **Safe mode**: this mode was meant to kick in whenever the star sensors could not be used to estimate the attitude, e.g. when their field of view was obstructed, and kept the spacecraft in a safe orientation with respect to the Sun. It makes use of the 12 coarse Sun sensors placed all over the spacecraft to estimate the Sun direction and keep the spacecraft rotating in order to maintain proper thermal management. This mode could transition between two states: Controlled Roll–Yaw and Eclipse state. The first would be active as long as at least one sun sensor was illuminated and first oriented the spacecraft such that the sun direction was perpendicular to the z-axis, to ensure maximum power generation, and then kept its angular velocity with a component along the z-axis, to ensure proper thermal management, and a component along the Sun direction to reduce the accumulation of momentum on the reaction wheels due to external torques. The second, as the name suggests, would kick in whenever the spacecraft went into eclipse and none of the Sun sensors gave out a strong enough signal. It relied only on the measurements of the gyroscopes and used the reaction wheels not to keep the spacecraft spinning about an axis, but just to cancel out the gyroscopic effect of the wheels, letting the disturbance torques take over. This was done to save power as the reaction wheels would not counteract the external torques during the entirety of the eclipse.

- **Fine pointing mode**.: this is the main mode, the one which achieved the precise pointing requirement needed by the payload, relying on the much more precise star trackers. Due to the varied nature of the payload, this mode achieved many pointing profiles, which are reported in table 18:

Attitude profile	Description
LDEX RAM	1 Rev/Orbit with +Z along neg. orbit normal
NMS	Survey Same as LDEX RAM but with arbitrary yaw angle
NMS	Ion Scan NMS ecliptic South with Panel 1 to Sun
UVS Limb (Backward)	UVS Tel. backward limb with +Z neg. orbit normal
UVS Limb (Forward)	UVS Tel. forward limb with +Z orbit normal
UVS Limb (North)	UVS Tel. North limb with Panel 1/5 normal to Sun
UVS Limb (South)	UVS Tel. South limb with Panel 1/5 normal to Sun
UVS Occultation	UVS Sol. View on Sun with +Z neg. orbit normal
UVS Sodium Tail	UVS Tel. along anti-Sun with +Z neg. orbit normal
COMM (Orbit Normal)	MGA toward Earth with +Z pos. orbit normal
COMM (Anti Orbit Normal)	MGA toward Earth with +Z neg. orbit normal
LLCD RAM	LDEX Ram with -60 deg offset about +Z
LLCD Earth	Same as COMM (Anti Orbit Normal)
Delta-V	Custom designed for maneuver attitude
Thermal Conditioning	+Z ecliptic normal and fixed rotation about Z-axis
Special Attitude	Custom designed for instrument needs

Table 18: Attitude profiles [17]

5.6 Pointing Budget

LADEE's pointing requirements varied significantly across its operational phases, driven by the active subsystem's needs. During Nominal Science, payload objectives were paramount; the UVS instrument required high pointing accuracy (better than 0.2° APE) and knowledge (better than 0.15° AKE), coupled with high stability for its remote sensing of the exosphere, while the LDEX and NMS instruments needed specific orientations, typically aligned with the velocity vector (Ram-pointing), for direct dust and gas measurements. This primary science pointing was periodically adjusted for dedicated high-rate communications phases, which demanded precise MGA alignment with Earth, representing the most critical pointing constraint for the TMTC subsystem. Outside these MGA contacts, essential command and telemetry relied on the less directive omnidirectional antennas, especially during survival or transitional states. In Safe Mode survival dictated the attitude: pointing was based on coarse sun sensors, achieving only $\pm 2°$ knowledge. Simultaneously, a controlled z-axis spin at approximately $0.15\,°\,s^{-1}$, complemented by a $0.1\,°\,s^{-1}$ Sun-vector spin for momentum management. During Maneuvering phases, accurate thrust vector alignment was the sole pointing driver, with payloads in safety and communications typically limited to the omnidirectional antennas. Finally, during eclipse, with no solar input, pointing was minimal, as EPS focused entirely on power conservation and TCS aimed to prevent components from reaching survival temperature limits. In the following table is reported what it was mentioned from the previous description.

Subsystem	Nominal Science	Safe Mode	Maneuvering	Communications	Eclipse Mode
P/L (UVS focus)	APE < 0.2°; AKE < 0.15°	N/A	N/A	N/A	N/A
P/L (LDEX/NMS)	Ram / Off-Ram / Ion Scan Orientations	N/A	N/A	N/A	N/A
TMTC (MGA)	-	-	-	Earth Pointing	-
Spacecraft Bus	Varies per P/L or Comm need	Z perp. Sun Target (AKE: ∼2°)	Thrust Vector Alignment	Earth Pointing (MGA)	N/A
Rates	Low (for stability)	Z-Spin: $0.15\,°\,s^{-1}$; Sun-Spin: $0.1\,°\,s^{-1}$	Low during coast	Low	N/A
EPS	-	Sun Pointing for Power	-	-	Power Minimize
TCS	-	Z-Spin for Thermal	-	-	Thermal Control

Table 19: LADEE Pointing Budget

5.7 AOCS Reverse Sizing

5.7.1 Inertia Model Derivation for LADEE Spacecraft (Wet Configuration)

Given the limited publicly available data regarding the LADEE spacecraft's mass properties, this study develops a simplified yet physically representative inertia model for the fuel-loaded ("wet") configuration. These calculations were essential to enable precise attitude dynamics simulations and disturbance analysis during the mission's science phase. The principal moments of inertia for the uniform mass distribution (prior to center-of-mass adjustments) were calculated as:

$$I_{xx} = I_{yy} = 484.63\,\mathrm{kg\,m^2} \tag{22}$$

$$I_{zz} = 227.51\,\mathrm{kg\,m^2} \tag{23}$$

A vertical center-of-mass offset of $-5.5\,\mathrm{cm}$ along the Z-axis was incorporated, corresponding to the fuel-loaded configuration's mass distribution as described in mission documentation. Application of the parallel axis theorem yielded the final adjusted inertia tensor:

$$I_{xx} = I_{yy} = 485.79\,\mathrm{kg\,m^2} \quad (\Delta = +1.16\,\mathrm{kg\,m^2}) \tag{24}$$

$$I_{zz} = 227.51\,\mathrm{kg\,m^2} \quad (\text{unchanged}) \tag{25}$$

These refined inertia values provide the necessary inputs for disturbance modeling and attitude control analysis, particularly for studying the spacecraft's response to external torques during its operations. The spacecraft structure was modeled as two primary components:

- A main octagonal body (mass = 303 kg, average radius = 0.925 m, height = 2.0 m)
- A tapered upper section (mass = 80 kg, lower radius = 0.925 m, upper radius = 0.5 m, height = 0.37 m)

5.7.2 Disturbance Torque Analysis

The disturbance analysis focused primarily on the science phase during lunar orbit, where environmental torques are well characterized. The model's flexible structure permits extension to the commissioning phase through appropriate orbital parameter adjustments. During the science phase, LADEE operated in a near-circular orbit at 50 km altitude with a 157° inclination, optimized to minimize eclipse periods. The two dominant disturbance torques were solar radiation pressure (SRP), arising from the spacecraft's large surface, and gravity gradient effects due to the spacecraft's dimensions at low orbital altitude. Aerodynamic drag was confirmed negligible due to the Moon's exosphere's low atmospheric densities. Similarly, magnetic torques were insignificant due to the Moon's lack of a global magnetic field, with only localized crustal magnetism, orders of magnitude too weak to affect spacecraft attitude. The torque estimation used a Simulink® model that simulated spacecraft attitude dynamics using verified orbital parameters from mission data. The simulation maintained fixed attitude conditions to isolate environmental disturbance effects, simplifying control system dynamics to focus specifically on torque characterization.

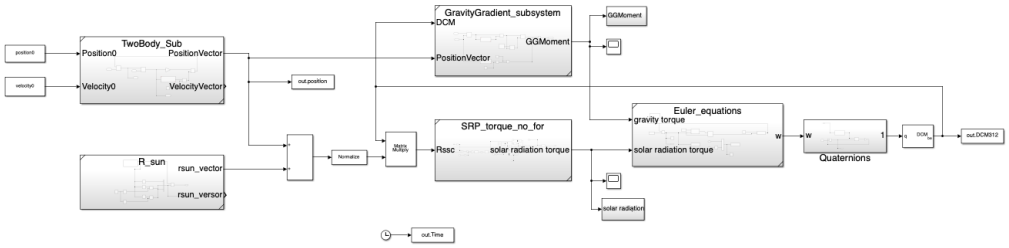


Figure 11: Simulink model.

To further validate the simulation results, mean values of the two dominant torques were analytically evaluated using the following equations:

$$T_{gg} = \frac{3\mu}{2R^3}(I_{\max} - I_{\min}) \sin 2\theta \qquad T_{\text{SRP}} = \frac{F_s}{c} A(1+q) \cos I (c_{\text{sp}} - c_g) \tag{26}$$

The parameters in these equations were selected to represent worst-case conditions:

- Sun incidence angle $I = 0°$;
- Center-of-pressure offset $(c_{\text{sp}} - c_g) = 0.0055\,\text{m}$;
- Reflectivity coefficient $q = 0.3$;
- Attitude angle $\theta = 270° - i$.

The solar radiation pressure torque was computed at each time step, along with the gravity gradient torque. During the mission, also a 4 h lunar eclipse occurred on April 14th, which verified the spacecraft's ability to withstand temporary loss of solar power. The gravity gradient torque remains non-negligible because the spacecraft attitude does not correspond to an equilibrium position ($\theta \neq 0°$). The analytical results from Equations 36 show a good agreement with the simulation outputs, as demonstrated in Table 20.

Disturbance	**Formula** $[N \cdot m]$	**Simulation mean** $[N \cdot m]$	**Simulation max** $[N \cdot m]$
SRP torque	$1.331 \cdot 10^{-6}$	$3.364 \cdot 10^{-6}$	$3.468 \cdot 10^{-6}$
GG torque	$2.393 \cdot 10^{-4}$	$1.568 \cdot 10^{-4}$	$2.394 \cdot 10^{-4}$
Total	$2.407 \cdot 10^{-4}$	$1.602 \cdot 10^{-4}$	$2.428 \cdot 10^{-4}$

Table 20: Disturbances overview.

5.7.3 Actuators Reverse Sizing

For the reverse sizing of the reaction wheels and secondary thrusters, the following topics were considered: momentum build-up due to disturbances per orbits and fastest or most critical slew maneuver performed throughout the mission.
For the values of the gravity-gradient torque, simulation-based literature [**Sun-Safe˙controller**] will be used since it provides more accurate information about the disturbances. The maximum torque reported for the gravity gradient is $6.5 \cdot 10^{-4}\text{N} \cdot \text{m}$.

5.7.4 Reaction Wheels

In this analysis, four reaction wheels arranged in a pyramidal configuration are used. Since no specific limit on the maximum allowable angular rate to ensure payload safety was provided, a conservative value of $\dot{\theta}_{max} = 0.5\,^\circ\,\text{s}^{-1}$ was assumed. Additional properties include:

- Maximum momentum storage per RW = 1.1 N m s
- Maximum torque per RW = 0.03 N m

Since the disturbance torque is mostly on the x-axis, 2 reaction wheels will be need to be de-saturated after n orbits, where n can be obtained from:

$$n = \frac{2h_{rw,max}}{h_D} = \frac{2h_{rw,max}}{2T_D \cdot t_{orbit}} \tag{27}$$

in this equation, a margin of 2 was applied on the disturbances torque. de-saturation will occur after **n=2.7** orbits, or 305 min.
Next, the critical slew maneuver, associated with sun-safe mode entry, will be considered. This mode requires orienting sensitive equipment away from the sun, potentially involving a worst-case $\theta_{slew} = 180°$ reorientation if initially sun-facing. As applying maximum reaction wheel torque would yield an angular rate exceeding the allowed limit, this rate was fixed instead. From the fixed rate, the slew time and the necessary maneuver torque were determined:

$$T_{RWs} = \frac{4\theta_{slew} I_{max}}{t_{slew}^2}$$

where $t_{slew} = \frac{\theta_{slew}}{\dot{\theta}_{max}}$. The slew time is 360 s and the used torque is 0.047 N m

5.7.5 Thrusters

There are four 22 N thrusters that have an inclination of 45° and an arm L of 0.8 m around the x-axis. For the aforementioned slew maneuver, two thruster will be considered with their component in the radial direction, such that, $F_{thrusters} = 2 \cdot 22\,\text{N} \cdot sin(45°)$. The thrust profile to consider will then be acceleration+coast+deceleration. The acceleration time is then:

$$t_{acc} = \frac{\dot{\theta}_{max} I_{max}}{F_{thrusters} L}$$

The resulting acceleration time is too small ($t_{acc} < 1\,\text{s}$), thus, an acceleration time of 1 s is considered. This makes the required thruster force 5.3 N, which would result in a coasting time of 359 s.
The thrusters will also be used for de-saturating the RWs, the time for de-saturating can be obtained from:

$$t_{des} = \frac{h_{RW,max}}{F_{thrusters} L}$$

Using two thrusters, the de-saturation time is 0.08 s, which is too small, thus a value of $t_{des} = 5\,\text{s}$ will be imposed, resulting in a thruster actual force of 0.55 N.

5.7.6 Mass and Power Budget

The thruster would be used in some slew maneuvers and most importantly in de-saturating the RWs. However, the slew phase frequency is not well known since it only happens in case of technical issues, which is compared to the frequency of de-saturation would be very small, so only the de-saturation fuel will be taken into account:

$$m_p = \frac{t_{des,tot} F_{th,des}}{I_{sp} g_0}$$

Where $t_{des,tot} = t_{des} \frac{n_{orb,tot}}{n_{orb,des}}$.
The required propellant mass is 0.65 kg. The power required for the RWs can be obtained from the datasheet [41], where the steady state operation consumes less than 9 W, and considering the very small ratio between the required toruqe to counteract the disturbances and the maximum torque, a value of 4 W will be considered for each RW, which means the total consumption is around 16 W.

6 Thermal control subsystem

Like any other space mission, LADEE's mission required a TCS to maintain the temperature of its components within the safe and operational ranges. This was critical in the s/c design, since the scientific instruments had requirements on the temperature operation range.
The LADEE TCS was primarily passive in nature, supported by minimal active elements, used only when necessary to maintain survival temperatures. The effectiveness of this largely passive strategy was enabled by LADEE's nearly symmetrical spacecraft configuration, symmetric solar array placement, and spin-stabilized attitude control [**Sun-Safe˙controller**]. These design choices contributed to an almost uniform thermal exposure across the spacecraft's surface, minimizing temperature gradients and significantly reducing the need for active thermal regulation or the presence of distinct "hot" and "cold" sides.

6.1 TCS components

LADEE adopted a simple passive choice of components for thermal management that are charactrized as following [22]:

6.1.1 Passive components

- **MLI**: A germanium-black Kapton MLI was used to cover the exterior parts of the NMS. It was also used to wrap the the UVS and LDEX. This film is charactrized by α/ε value of 0.93 [19]. It's transparent in terms of Radio Frequencies.
- **Thermal Coatings**: The external parts of the UVS and the NMS's MEB external parts were coated with a silverized teflon tape, which has α/ε value of 0.13 to 0.16 and is designed to be directly exposed to the [20]. The hemi-spherical part at the detector of the LDEX is also coated with rhodium.
- **Radiators**: a radiator module was mounted on top of the s/c, it consisted of a high-emissivity panel facing the deep space. Attached to it are critical components such as the UVS and LDEX assembly, avionics board, the battery and part of the communications subsystem.
- **Conductive connections**: conductive heat mounting were used to mount the components. Cho-Therm 1671 gaskets were used in mounting the UVS's spectrometer to its radiator and also mount instrument's solar viewer and telescope to the side of the assembly. This material has a fiberglass enformcement and a thermal conductivity of $2.6\frac{\mathrm{W}}{\mathrm{mK}}$ with electrical insulation [11].
- **Isolative connections**: Isolating materials were used to make some components thermally independent of the s/c. For example, titanium flexure mounts were used to isolate LDEX from the s/c and mechanically support it.

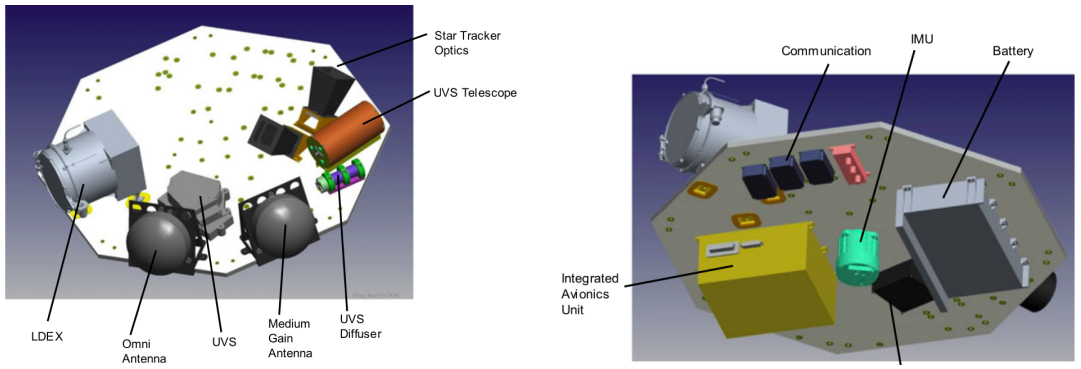
6.1.2 Active Components

The use of active components was limited to using heaters. Electrically powered resistive heaters were installed in temperature-critical components such as the battery and scientific instruments. for NMS, it was specified that Kapton foil heaters were used.

6.2 TC Design and Justification

At the subsystem level, LADEE's TCS combines passive methods, such as MLIs, with active heaters for critical components. The top deck, acting as a radiator, contains heaters to prevent excessive cooling.

Active heaters are also used on the instruments, LLCD optics, propulsion system and lines, batteries, transponder, and upper radiator panel, all controlled by spacecraft avionics. Thermal challenges are most pronounced during telemetry orbits and lunar noon, the hottest periods. There are no significant cold cases during operations, though prior to LOI, LADEE can withstand up to a 1-hour eclipse before exceeding low-temperature limits [52]. The TCS design was shaped by strict pointing requirements and autonomous lunar operations [22]. Passive elements minimized thermal cycling during repeated lunar day-night transitions, reducing structural distortion risks. Instrument placement was optimized to ensure uniform thermal conditions, with payloads grouped on adjacent panels (Fig. 12).



(a) Radiator assembly top view.

(b) Radiator assembly bottom view.

Figure 12: Radiator assembly.

Active heaters were limited to essential thermal nodes. Radiators, mounted on deep space-facing sides, efficiently expelled excess heat. Body-mounted solar panels, when Sun-facing, dictated much of the spacecraft's internal thermal profile. LLCD operation was managed through combinations of altitude, attitude, and duration to fine-tune thermal conditions [56]. MLI was selected to minimize contamination, ensure thermal decoupling between the PLM and SVM, and maintain surface conductivity. Coatings and embossed Kapton enhanced thermal performance and reduced mass, contamination, and outgassing risks.

6.3 Thermal Ranges and Phases

In order to succeed in the mission objectives each component on the spacecraft had to be mainteined in a specific range of temperatures, as described in Table 21.

In the table, the data regarding sensors and RWs are taken from datasheet [2] [41] [3] [60] [7], while the others are typical thermal requirements found in literature [22] [68]. Also some clarifications have to be made: the two ranges for the batteries represent the limits for charge and discharge phases. The temperature ranges for PLM and SVM depend on the specific mission and there is no valid general data. The Sun Sensors are also qualified for excurses of $-145\,°C$ to $120\,°C$. The UVS has also some specific technical constraints for

Component	**Minimum T** [°C]	**Maximum T** [°C]
Batteries	0/-20	+45/+60
RW	-30	+60
IMU	-40	+70
Star Tracker	-30	+60
Propellant	+7	+40
Antennas	-40	+80
Sun Sensors	-50	+80
LDEX	-25	+50
NMS	-20	+40
UVS	-20	+45
LLCD	-30	+50

Table 21: Operational temperatures of S/C components.

which its temperature has to be manteined at least 10 °C above the adjacent spacecraft surfaces, tipically <20 °C. In the literature regarding the temperature ranges of MMH and NTO [68] the operative temperature range is not specified so the used values are obtained applying a 50% margin to the reported nominal temperature range observed in past missions. The following Table 22 reports a description of heat sources and sinks for each phase of the mission:

Phase	Description	Heat Source	Heat Sink
LEOP	Launch and Early Orbit Phase	Sun, Earth albedo, Earth IR, internal power	Deep Space
TTP	Transfer Trajectory Phase	Sun, internal power	Deep Space
Lunar Orbit Phase	Science operations in low lunar orbit	Sun, Moon albedo, internal power	Deep Space

Table 22: Thermal characteristics of LADEE mission phases.

6.4 Units Specifically Controlled

LADEE's structural design, constrained by the MCSB, which included body-mounted solar panels and no deployable radiators, limited the spacecraft's ability to actively manage heat rejection and created a strong thermal coupling between the whole structure and the solar arrays. As a result, localized insulation and heater control were essential to maintain each subsystem within its required temperature range [54]. Both survival and operational heaters, controlled by spacecraft avionics, ensured thermal compliance during cruise, observation, and standby phases; They were implemented because they are adaptable to different geometries and easy to control, making them ideal for ensuring independent regulation.

6.4.1 Payload

LADEE's scientific instruments employed a mix of passive and active thermal control strategies, with heaters and MLI forming a common foundation across all payloads. The NMS implemented this standard approach, combining passive elements with controlled heaters. The LDEX was thermally isolated using titanium flexures and featured a rhodium-coated surface to limit solar heating, relying also on a survival heater. The UVS required greater precision and therefore integrated a thermoelectric cooler governed by a PID loop, with Cho-Therm gaskets enhancing thermal conduction at critical points. Finally, the LLCD optics, sensitive to thermal extremes, depended on both heaters and insulation to ensure performance stability across mission phases.

6.4.2 Propulsion System

For this subsystem, the requirements are derived from the propellant temperature range defined in the table of the previous section [21]. To make sure the constraint are respected, the propulsion lines and thruster components were insulated using MLI and actively heated.

6.4.3 TMTC

This subsystem likely employed the same thermal control strategies used by the others, namely MLI and heaters. However, additional antenna-specific solutions may also have been implemented. In particular, a plausible approach would be the application of high-reflectivity white coatings on Sun-exposed antenna surfaces to minimize solar absorption.

6.4.4 AOCS

AOCS components were also thermally managed using a mix of passive insulation and active heating. Star trackers and IMUs, which are highly sensitive to thermal gradients and bias drift, were thermally

isolated and, where needed, regulated with controlled heaters to maintain tight temperature stability. Reaction wheels, being internally dissipative, were mounted on conductive baseplates with thermal paths to the spacecraft structure, and were wrapped in MLI to limit temperature oscillations. Survival heaters were implemented to maintain lubricant performance and mechanical integrity during non-operational phases.

6.4.5 EPS

For LADEE's electrical system, it is highly likely that the thermal control approach mirrored that of the scientific payloads: electronics units were thermally isolated with MLI and supported by heaters. A critical aspect of LADEE's TCS involves the thermal management of the body-mounted solar panels. Due to the limited publicly available information about the actual thermal design, a possible solution is proposed based on consolidated spacecraft thermal control technologies [30]. To dissipate the heat accumulated by the illuminated solar panels, heat pipes could be used to transfer the excessive heat to the main radiator. However, since the radiator already hosts instruments and sensors that contribute to its thermal load (and may reduce its efficiency), it may be advantageous to also route some of the heat to the non-illuminated solar panels. These shaded panels, acting as additional radiators, would help distribute the thermal load more evenly across the spacecraft surfaces. This solution should be complemented with MLI to prevent overheating of thermally sensitive areas inside the MCSB.

6.5 Reverse sizing

In the following section we are going to model LADEE's thermal behaviour throughout the mission to understand the rationale behind the design choices of the TCS.

6.5.1 Model constraints

A preliminary step in the analysis is identifying the allowed temperature range and when extreme cold and hot condition happen. As listed in Table 21 each component has its own range of operating temperatures, but it can be seen that the most demanding is dictated by the propellant, so we chose to maintain the temperature in the slightly less conservative range of 0°C and +40°C, as it would satisfy the other constraints as well. During the whole mission the coldest temperature would be reached during eclipse before Lunar Orbit Insertion, as the high altitude ($\sim$ 4000km) would drastically decrease the infrared radiaton received; but as stated in [52] LADEE could only withstand about an hour in that condition before the temperature reached critical values, so applying our steady-state model (i.e. temperature after infinite time) would not be representative of the real behaviour; so we identified the cold case to happen during eclipse in the commissioning orbit with an altitude of 200km and a minimum internal power generation of 84W[1]. For the hot case the choise was between the science orbit and the first Earth-centered phasing loop, with Sun exposure in both. Quick calculations of the incoming radiation power received reveal it to be higher in the science orbit, so that was our choice for the hot case, with the addition of a maximum internal power generation of 196W[2].

6.5.2 Single-node model

The model we first used is a very simple one, where we modelled the entire spacecraft as a sphere of equivalent surface area as previously stated, and computed the equilibrium temperature, not considering any transient behaviour. The surface area was computed assuming the spacecraft to be an octagonal prism 2.37 m high and 1.85 m wide, for a total surface area of 20.2 m^2. The side faces are

[1] From [14] we see a minimum power consumption of 3A at 28V
[2] Again from [14] we find 7A at 28V

covered in solar panels while the top and bottom panels' material is not available in the literature, so we went with aluminium as it is lighweight and has good thermal conductivity, covered with white paint for its low absorptivity and high emissivity, ideal for a radiator. From [31] we obtain $\varepsilon = 0.82$ and $\alpha = 0.78$ for the solar cells and $\varepsilon = 0.92$ and $\alpha = 0.17$ for the radiator; which we then average with respect to the percentage of total area occupied by each one to obtain the final values for the sphere as a whole, which yields $\varepsilon = 0.0.75$ and $\alpha = 0.62$.

6.5.3 Heat sources

The next step is computing the heat sources, which come from: the components of the spacecraft, the Sun, the Moon's albedo and the Moon's infrared radiaton. We didn't include infrared and albedo radiation from Earth as they are negligible due to the distance from it. The incoming heat Q can be calculated with the following formulas:

$$Q_{sun} = A_{cross}^{sun} \alpha_{sc} q_0 \qquad Q_{albedo} = A_{cross}^{planet} a \alpha_{sc} q_0 \left(\frac{R_{moon}}{r_{orbit}} \right)^2 \tag{28}$$

$$Q_{IR} = A_{cross}^{planet} \varepsilon_{sc} \sigma \varepsilon_{moon} T_{moon}^4 \left(\frac{R_{moon}}{r_{orbit}} \right)^2 \tag{29}$$

Where q_0 is the solar flux 1au from the Sun, a is the planetary albedo, A_{cross}^{sun} is the cross-sectional area of the sphere and A_{cross}^{planet} has been computed with the following formula based on the orbital altitude h:

$$A_{cross}^{planet} = A_{sc} \frac{1}{2} \left(1 - \frac{\sqrt{\left(\frac{h}{R_{moon}} \right)^2 + \frac{2h}{R_{moon}}}}{1 + \frac{h}{R_{moon}}} \right) \tag{30}$$

The results are reported in Table 23

	Q_{sun}	Q_{albedo}	Q_{IR}	Q_{int}
Hot case	4251W	836W	3192W	196W
Cold case	0W	0W	1980W	84W

Table 23: Heat sources

6.5.4 Hot case

By imposing heat balance we can compute the temperature reached in the hot case, which is given by:

$$T_{sc} = \left(\frac{Q_{sun} + Q_{albedo} + Q_{IR} + Q_{int}}{\sigma \varepsilon_{sc} A_{space}} + T_{space}^4 \right)^{\frac{1}{4}} \tag{31}$$

where A_{space}, the area that radiates to deep space, is just the total area of the spacecraft minus the area that receives radiation from the Moon, and the temperature of deep space T_{space} is set to 3K; this yields a temperature of 82°C, way above the maximum required. To fix it we can try to compute the area of a radiator needed to bring the equilibrium temperature down to 25°C, which is the maximum temperature allowed with a margin of 15°C. The area is computed with the following formula:

$$A_{rad} = \frac{Q_{sun} + Q_{albedo} + Q_{IR} + Q_{int} - \sigma \varepsilon_{sc} A_{space} \left(T_{max}^4 - T_{space}^4 \right)}{\sigma \varepsilon_{sc} A_{space} \left(T_{max}^4 - T_{space}^4 \right)} \tag{32}$$

where the radiator's emissivity was again chosen to be that of a white paint as stated before; this yields a surface area of 10.6m^2. Since LADEE didn't make use of deployable radiator and the area we obtained is less than the total area of the spacecraft it is reasonable to think of converting part of the solar panels into radiating surfaces, even though this would mean converting most of the solar cells into radiators, thus compromising the generation of electrical power.

6.5.5 Cold case

The temperature for the cold case can be calculated with the same formula used for the hot case by also taking into account the power emitted from the radiator whose area was computed above, obtaining the formula:

$$T_{sc} = \left(\frac{Q_{sun} + Q_{albedo} + Q_{IR} + Q_{int}}{\sigma \left(\varepsilon_{sc} A_{space} + \varepsilon_{rad} A_{rad} \right)} + T_{space}^4 \right)^{\frac{1}{4}} \tag{33}$$

yielding a temperature of -68°C, way below any reasonable threshold. To increase the temperature to 15°C we can think of implementing electrical heaters to generate more heat. By integrating such heat in the energy balance we find that:

$$Q_{heat} = \sigma \left(\varepsilon_{sc} A_{space} + \varepsilon_{rad} A_{rad} \right) \left(T_{min}^4 - T_{space}^4 \right) - Q_{IR} - Q_{int} \tag{34}$$

resulting in 7.4 kW, way too much for the power system to generate and provide, especially after converting part of the solar cells to radiator.

6.5.6 Single-node results

These results show how the single-node approach has its very big limitations and cannot be applyed to model the complex thermal behaviour of an entire spacecraft. As previously mentioned LADEE implemented a more complex TCS, with localised heaters and thermal interfaces to control the temperature of each component, so it is clear that reducing everything down to a single component is not satisfactory.

6.5.7 Advanced model

In the hope of obtaining better results, we developed an 18-node steady-state thermal model in MATLAB. The spacecraft was approximated as an octagonal prism, and the Sun and Moon were considered to orbit in the middle plane between the top and bottom panels. Nodes 1–8 represent the external side panels ($\varepsilon = 0.8$ $\alpha = 0.7$), nodes 9 and 10 the top and bottom panels ($\varepsilon = 0.9$ $\alpha = 0.1$), and nodes 11–18 the internal parts of the side panels ($\varepsilon = 0.1$ $\alpha = 0.2$). The external nodes (1–10) can exchange heat via radiation with the Sun, the Moon (IR and albedo), and deep space. Furthermore, side panel to side panel and side panel to top/bottom panel conductance is considered. The internal nodes are insulated from the external side nodes (1–8) with MLI and can exchange heat among themselves and with the top and bottom panels by conduction and internal radiation.
The equilibrium temperature in the cold case was -17°C, while in the hot case it was 45°C.
This model demonstrates that the radiator panel is effective at rejecting heat because it is perpendicular to the solar panels and is almost always facing deep space. It also demonstrates the effectiveness of the MLI insulation in keeping the internal temperature higher. However, this model is still far from a complete representation of the actual thermal behaviour, for which an extremely complex model would be needed.

7 EPS Architecture

LADEE's Electrical Power System (EPS) reflected the design criteria of the whole mission as a simple, low-cost and reliable system. The system's components are mainly Commercial Off-The-Shelf or based on previous missons and consisted of an array of 30 body-mounted solar panels, a single 24 A h lithium-ion battery and an Integrated Avionics Unit (IAU) that contained all the power electronics. Figure 19 shows a diagram of the system; it can be noted that all the loads can be switched on and off the unregulated bus independently, while the solar panels are routed through 12 switches; furthermore the common ground for all of the electronics was the s/c structure itself, through wich all of the return currents would be routed.

7.1 Power sources

The main source of electrical power are the 30 body-mounted silicon-cell solar panels that cover the sides of the octagonal shape (except for the 2 side panels where the NMS and LLCD are located). Each panel produces around 1 A [25], considering the maximum power generated by the system was about 339.7 W while the required one was 295 W[25]. The 30 panels could be switched on and off the bus through 12 switches, so they were divided in 12 groups to ensure that a failure in any one of the switches would still allow for uniform power generation, as it can be seen in figure 14. In order to provide power during eclipses and assist the solar panels during peak loads the system relied on a single lithium-ion battery. The output voltage ranged from 22 to 34 V depending on the state of the battery, for an average of 28 V, thus yielding a capacity of about 691 W h [22].

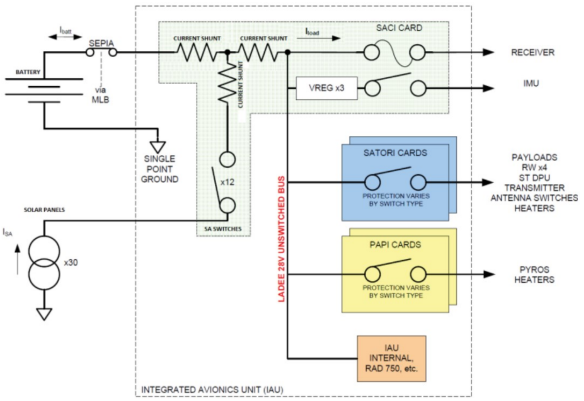


Figure 13: LADEE's Electrical Power System [25]

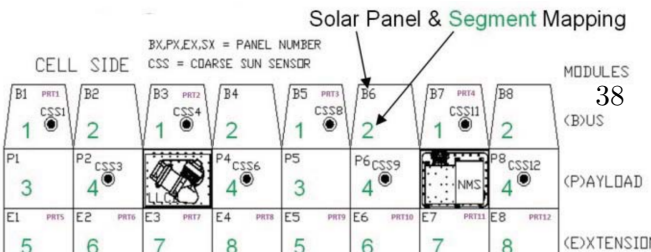
7.2 Power Distribution

LADEE EPS architecture did not use a dedicated power distribution board. Instead, the battery bus provided the voltage unregulated to each components [25]. The only contol was applied in order to prevent excessive charging/discharging of the battery. This was done via interfacing the power generated by the solar panels and the batteries through SACI card, which is managed by the the IAU, a COS 8-slot 3U cPCI manufactured by Broadreach Engineering [34]. Each component of the s/c was then responsible to step down or step up the voltage, based on its needs and the provided voltage, to meet its power requirements. The IAU was responsible for monitoring the power feeding lines and switching them on and off based on the mission needs and events.

7.3 Design Justification

The use of body-mounted solar panels eliminated the need for deployement mechanism, which suits the mission low cost & low complexity. The arrangement of the solar panels on the octagonal shape of the s/c allowed efficient thermal management through attitude control [64], which in turn also reduced the complexity of the TCS.

LADEE's power requirement was 295 W, with solar panels' average output of 310 W [25]. The typical power requirements for LADEE's



modes -excluding contingency margins- are below this value, as reported in [**??**], with nearly 80 % margin during the science operations. This covered contingencies (battery charging, inefficient orientations, degradation). However, other modes showed little to no margin, which could be due to our conservative estimation for the power requirements or due to the mission's budget, time and mass constraints. The Li-Ion battery choice was well suited for the mission's long life time and its high battery capacity allowed for conserving mass and volume, which was crucial for LADEE's low budget. The battery's capacity of around 691 W h [22] also allows to operate the s/c for more than 3 hours during prolonged eclipses. Using unregulated power supply eliminated having a single-point of failure in the power distribution. Using a single current-shunt measurement was to reduce costs. In addition, the entire EPS, with the exception of the wiring harness, was based on COTS. All of this is in-line with mission's low-cost and rapid approach.

7.4 Operational profiles and power budget

To size the LADEE EPS system, a power budget breakdown was conducted for each subsystem across key operational phases, guided by the operational profiles detailed in Figure 15:

Subsystem	Science	Technical Dem. (LLCD ops)	Maneuvre (OMM/TCM/LOI)	Comm. (Dedicated comunication Pass)	Eclipse
AOCS	Science Pointing (RWs, ST, IMU,SS)	Precise Earth Ponting (RWs,ST,IMU,SS)	Thruster Pointing Control (OCS/RCS,ST,IMU,SS)	Earth Pointing (RWs,ST,IMU,SS)	Stable Pointing (Minimal) (RWs,IMU,ST)
TCS	✓	✓	✓	✓	✓ (Survival Heaters-High Load)
TT&C	✓	✓ (LLCD Support)	✓	✓	✓ (Minimal Health Telemetry)
OBDH	✓ (Science Data Handling)	✓ (LLCD Data Handing)	✓ (Maneuvre Sequencing)	✓	✓
Propulsive	RCS for OMMs / Momentum Dumps	X	✓ (OCS and/or RCS)	X	X
EPS	✓	✓	✓	✓	✓(Battery Discharge)
Payload-Science	✓	X	X	X	X
Payload-LLCD	X	✓	X	X	X
Energy Source	Solar Panels	Solar Panels+Battery	Solar Panels (Battery for long/high-power burns if needed)	Solar Panels	Battery

Figure 15: Operational Profiles (which subsystems are active in each phase)

During Sunlit Operational Phases (science, comm and typical maneuvres), LADEE primarily relies on its solar panels. These phases are explained as follows:

- **Manoeuvre phases:** During typical manoeuvre phases (OMMs, TCMs), the Propulsive system (OCS or RCS) is the main active element, with AOCS controlling the burn. Both science and LLCD payloads are inactive to ensure stability and dedicate resources.
- **Comm phase:** During dedicated high-volume communication passes, TMTC operates at high power for S-Band transmission, with AOCS ensuring accurate Earth-pointing of the MGA. Both science and LLCD payloads are typically inactive to maximize power for the communication link.
- **Science phase:** The AOCS maintains precise pointing (ram, limb, or occultation) using reaction wheels and star trackers, enabling the *Payload* (NMS, UVS, LDEX, typically time-delayed due to power constraints) to gather data. TMTC facilitates telemetry downlink and command reception, while the Propulsive system (RCS) executes periodic OMMs and momentum dumps.

The Technical Demo. phase, dedicated to LLCD operations, represents a high-power sunlit scenario, which required also precise Earth-pointing by AOCS. Due to the LLCD's significant power demand, this phase often necessitated supplementing solar panel power with battery energy. Consequently, primary science instruments are typically inactive. TMTC and OBDH support LLCD command, telemetry, and data handling. During eclipse, LADEE relies entirely on its battery. All non-essential systems, including science payloads, LLCD and propulsive system, are powered off. The TCS becomes a primary power consumer, with survival heaters engaged to protect critical components from low temperatures. AOCS maintains minimal stable pointing, while TMTC and OBDH operate at reduced levels for health monitoring and post-eclipse command reception. Core systems like OBDH and the EPS form a baseline power consumption and are considered always operational. The TCS actively manages thermal conditions throughout all phases, adapting its strategy based on the thermal environment and power availability, with its most intensive active heating occurring during eclipses.

Subsystem	**Science**	**Technical Dem.**	**Manoeuvre**	**Comm.**	**Eclipse**	**References**
PS	–	–	96.0W	–	–	[62][57][42]
Payload	41.0W	141.0W	–	–	–	[22] [12]
TMTC	25.0W	25.0W	25.0W	167.0W	25.0W	[33] [35]
AOCS	55.0W	55.0W	55.0W	55.0W	55.0W	[41] [3] [7] [60]
TCS	–	–	–	–	102.0W	[30]
Other systems	40 W	40 W	40 W	40 W	40 W	Assumed
Total	161 W	261 W	216 W	272 W	222 W	
Total +20%	193.2W	313.2W	259.2W	326.4W	266.4W	

Table 24: LADEE Power Budget by Operational Phase.

A detailed power breakdown at component level is not available in the provided documentation. The category "Other systems", with a constant power consumption of 40 W, includes essential bus functions such as OBDH, internal EPS consumption, and basic TCS operations. During sunlit phases without dedicated TCS power allocation, this value likely accounts for continuous thermal monitoring and minimal active heating, with the rest of the thermal control managed passively or via heat from other subsystems. A 20% power margin is applied to each mission phase to cover unlisted minor components and battery charging needs, which will be defined after power system sizing.

7.5 EPS Reverse Sizing

7.5.1 Solar Arrays

For the LADEE spacecraft, the Solar Arrays (SAs) are selected as the most suitable primary power source. The preliminary sizing of the Electrical Power System (EPS) begins with a summary of the power budget, reported in Table 24, based on the operational profiles defined for the mission. This analysis identifies the Telecommunication Phase, during which LADEE utilizes the medium gain antenna, as the period of maximum power demand. This scenario is considered as the sizing case, with a peak power requirement of $P_{\max} = 326.4\,\mathrm{W}$ which includes a 20% engineering margin. The selected case represents a conservative worst-case scenario for sizing purposes, despite the fact that LADEE typically operated within a designed power range of 260–295 W [22]. The key parameters used for the solar array sizing analysis are summarized in Table 25.

Parameter	Symbol	Value
Sun irradiance	I_0	$1367.5\,\mathrm{W\,m^{-2}}$
Nominal lifetime	T_{lifetime}	0.6 year
Degradation per year	d_{py}	0.03
Sun incidence angle	θ	5°
Inherent degradation	I_d	0.7
BOL efficiency of single SA cell	ϵ_{BOL}	30%
Operating voltage of the system	V_{sys}	34 V
Operating voltage of a single SA cell	$V_{\text{cell,SA}}$	0.628 V
Area of single cell	A_{cell}	$22.3\,\mathrm{cm^2}$ - $27.5\,\mathrm{cm^2}$
Efficiency factor in daylight	X_d	0.85
Efficiency factor in eclipse	X_e	0.65

Table 25: Parameters chosen for SA sizing

The solar irradiance I_0 at the spacecraft, due to the small Moon-Earth distance, can be approximated by q_0, the solar flux at 1 AU. The LADEE mission lifetime is set to 0.6 years, including 186 days of science operations and 38 days for transfer and phasing. A sun incidence angle of $\theta = 5^\circ$ represents a near-worst-case scenario, reducing the effective illuminated area. The beginning-of-life (BOL) efficiency of the solar cells is 30% [23]. An inherent degradation factor $I_d = 0.7$ accounts for losses due to temperature, partial shadowing, and integration effects. The spacecraft employs a Direct Energy Transfer (DET) system, which maintains a fixed bus voltage and dissipates excess energy, offering a simpler but less efficient alternative to Peak Power Tracking (PPT). The power conversion efficiencies are $X_d = 0.85$ in sunlight and $X_e = 0.65$ in eclipse. With eclipse and sunlight durations of $T_e = 2940\,\mathrm{s}$ and $T_s = 3840\,\mathrm{s}$ respectively, the required solar array power is:

$$P_{SA} = \frac{\frac{PeTe}{Xe} + \frac{PsTs}{Xs}}{T_s} = 697.8\,\mathrm{W} \tag{35}$$

The power per unit area generated by the solar arrays (SAs) at the beginning of life (BOL) is calculated using the following equation: $P_{BOL} = I_0 \epsilon_{BOL} I_d \cos\theta \lambda_{45} \lambda_{30}$. In this expression, the geometric correction factors λ_{45} and λ_{30} are introduced to account for the average effective solar incidence angle on panels that are not perpendicular to the Sun's rays. These factors are computed using the following expression:

$$\lambda = \frac{1}{N} \sum_{i=1}^{N} \cos(\alpha_i)$$

where α_i is the incidence angle between the i-th panel and the direction of the incoming sunlight, and N is the number of inclined panels considered. This approach allows for an estimation of the average contribution to power generation from each group of panels, taking into account the structural orientation of the spacecraft. Using the parameters provided in Table 25, the initial power output per unit area is determined as: $P_{BOL} = 222.5\,\frac{\mathrm{W}}{\mathrm{m^2}}$. Over time, the performance of the solar arrays degrades due to several factors, such as radiation exposure, thermal cycling, and material corrosion. These are collectively quantified by the annual degradation rate, d_{py}. The cumulative degradation factor over the mission lifetime, T_{lifetime}, is computed as: $L_d = (1 - d_{py})^{T_{lifetime}} = 0.9815$. From this value, the end-of-life (EOL) power per unit area and the total required surface area of the solar arrays can be derived using:

$$P_{EOL} = P_{BOL} \cdot L_d = 218.39\,\frac{\mathrm{W}}{\mathrm{m^2}} \qquad A_{SA} = \frac{P_{SA}}{P_{EOL}} = 3.1952\,\mathrm{m^2} \tag{36}$$

The spacecraft features 22 panels on the Payload, Extension, and Single Stage Extension Modules, each using Type 1 solar cells of 22.3 cm^2. Additionally, 8 panels on the bus module employ Type 2 cells of 27.5 cm^2 area [22]. As the exact total surface area of the LADEE solar arrays and the number of cells per panel are not explicitly provided in the literature, assumptions are made based on geometry. It is estimated that 30% of the array area is from the top-mounted panels, with the remaining 70% on the bus. Given A_{cell}, A_{SA}, system voltage V_{sys}, and cell voltage V_{cell}, the number of required cells is:

$$N = ceil\left(\frac{A_{SA}}{A_{cell}}\right) \qquad N_{series} = ceil\left(\frac{V_{sys}}{V_{cell}}\right) \qquad N_{real} = ceil\left(\frac{N}{N_{series}}\right) N_{series} \tag{37}$$

This configuration ensures that the designed solar array provides the necessary power output while satisfying the voltage requirements of the spacecraft power system. The obtained results for both Type 1 and Type 2 cells can be summarized as reported in Table 26.

The actual voltage delivered by the system can then be evaluated as $V_{\text{real}} = N_{\text{series}} V_{\text{cell}}$, resulting in 34.54 V. The effective surface area of the solar arrays is computed as $A_{\text{SA,real}} = N_{\text{real},1} A_{\text{cell},1} + N_{\text{real},2} A_{\text{cell},2}$, yielding a value of 3.389 m^2. In this analysis, we considered the worst-case illumination scenario, where one of the two instruments mounted on the side of LADEE is exposed to sunlight.

	N	$N_{\textbf{series}}$	$N_{\textbf{real}}$
Type 1	349	55	385
Type 2	1003	55	1045

Table 26: Number of cells.

Consequently, only 8 lateral panels and 3 of the top (inclined) panels are assumed to be illuminated. To estimate the total actual solar array area, we computed the ratio r between the area of a single top trapezoidal panel and that of a single lateral rectangular panel from the data used in a previous assignments (H4). Using this ratio, we expressed the area of a single rectangular panel as: $A_{rett} = A_{\text{SA,real}}/(3 \cdot r + 8)$. The total area is then calculated as: $A_{tot} = (22 + 8 \cdot r)A_{rett} = 9.28\,m^2$. The power supplied in the worst case by the solar panels, at the beginning and end of life results in:

P_{SA}	P_{BoL}	P_{EoL}
697.8 W	754.1 W	740.1 W

Table 27: Power required and supplied at beginning and end of life.

The total mass of the solar panels can then be determined using the panel's specific power, obtained from datasheet specifications [24], where $SP_{\text{panel}} = 75\,W\,kg^{-1}$. Based on the beginning-of-life (BOL) power output, the resulting total mass of the panels is $m_{\text{panels}} = 27.53\,kg$. The discrepancy between the estimated solar array area and the actual value [3] ($\sim 13.05\,m^2$) arises primarily from the simplifications and assumptions made during sizing. These include idealized sun incidence angles, uniform illumination, and efficiency factors, which may not fully capture the spacecraft's real geometry, panel layout, and performances, leading to a better performing solar panels and consequently to a smaller necessary area.

7.5.2 Secondary Batteries

The LADEE design incorporates a single lithium-ion battery ($N_{\text{batt}} = 1$). For the battery sizing, an approach aligned with the state-of-the-art Li-ion battery technology available for space applications at the time was adopted, as detailed in [6]. While the study does not directly reference the LADEE mission, it presents a representative battery module and provides a realistic base for missions of the

[3] This value is based on geometrical assumptions, neglecting gaps between panels but accounting for the absence of solar panels on the lateral payload areas and the spacecraft's top and bottom surfaces.

same era in terms of technological maturity, modular configuration, and power management strategy. The relevant parameters, sourced from documentation, are summarized in Table 28.

Parameter	C_{cell} [W h]	E_v [W h dm^{-3}]	DoD [–]	η [–]	$V_{cellBatt}$ [V]	μ [–]
Value	5.5	233	0.5	0.8	3.6	0.85

Table 28: Parameters chosen for Battery sizing

The selected battery characteristics reflect its intended operational use, incorporating a relatively low Depth of Discharge to account for frequent cycling and to minimize long-term degradation. The key parameter for battery sizing is the total energy capacity, E_{batt}, which is determined using the corresponding formula provided below:

$$E_{batt} = \frac{\frac{P_e T_e}{X_e}}{N_{batt}\eta DoD} = 836.7692\,\text{W h} \tag{38}$$

The numerator of this equation represents the energy required by the system during periods when the solar arrays are unavailable. As with the solar array sizing, the battery sizing can be refined by determining the number of cells connected in series and in parallel, based on the characteristics of the selected battery cell model. Consequently, the energy capacity of a single string can be evaluated, accounting for the packing efficiency loss factor. This leads to the estimation of the total energy stored in a single battery string. Using the same formula from Eqs. 37 the number of battery cells in series results in: $N_{series} = 10$. To complete the sizing process and determine the total number of cells, the number of parallel strings must be evaluated as follows:

$$E_{string} = \mu C_{cell} \cdot (V_{cellBatt} \cdot N_{series}) = 168.3\,\text{W h} \qquad N_{parallel} = ceil\left(\frac{E_{batt}}{E_{string}}\right) + 1 = 6 \tag{39}$$

With the number of parallel strings now defined, the total number of cells in the battery can be readily computed as $N_{cells} = N_{parallel} \cdot N_{series} = 60$. This also enables the evaluation of the overall energy capacity and total electric charge of the battery module:

$$E_{batt,real} = N_{parallel} \cdot E_{string} = 1009.8\,\text{W h} \qquad C_{batt} = N_{parallel} \cdot \mu \cdot C_{cell} = 28.05\,\text{A h} \tag{40}$$

The battery capacity obtained in this study is higher than the actual LADEE's value 691 W h, this results can be attributed to conservative design choices. Specifically, a lower DoD (50%) and system efficiency factor ($\eta = 0.8$) were adopted to reflect modern design practices aimed at enhancing battery longevity and reliability under frequent cycling. The discrepancy is also accentuated by the high P_e value, chosen to represent the worst case eclipse scenario. These safety margins, combined with a realistic yet modular sizing approach, result in a larger energy reserve, ensuring mission robustness even under worst-case eclipse and peak load scenarios. The battery's *specific energy* and *volumetric energy density* are derived from the performance characteristics reported in [6]. According to this reference, the battery exhibits a specific energy of $SP_{\text{batt}} = 111\,\text{W h kg}^{-1}$ and a volumetric energy density of $\rho_E = 233\,\text{W h L}^{-1}$. These values reflect the adoption of high-efficiency Li-ion cells combined with compact electronics based on COTS components, optimized to minimize mass and volume in space environments.

$$M_{battery} = \frac{E_{batt,real}}{SP_{batt}} = 9.0973\,\text{kg} \qquad V_{battery} = \frac{E_{batt,real}}{E_d} = 4.3339\,\text{L} \tag{41}$$

8 OBDH Architecture

LADEE's OBDH architecture utilized a single-string centralized data handling center for operating the s/c and handling the commands, acquire data from the payload, transmitting and storing it, all using specific interfaces [8]. the chosen components were mostly COTS ones that also had a flight heritage. The IAU was an 8-slot 3U BroadReach board [22]. This board acted as the backbone to accomodate different chips and processors that controlled and operated LADEE, enabled interfacing different instruments and also managing the power supply for the s/c.
The first slot was occupied by a radiation hardened RAD750 microprocessor, it was the main OBC. Other two slots were AMOAB and DMOAB. AMOAB was mainly responsible for interfacing with analog data sources like the temperature sensors and the sun-sensors. and also managing SACI. DMOAB provided low speed communication with UVS, LDEX, TCTM subsystem, Propulsion subsystem, Star tracker, IMU and reaction wheels via RS-422 serial BUS. As well as high speed communication with the NMS and LLCD instruments through an LVDS BUS. The DMOAB was equipped with 1024 Mbytes of SDRAM [34], suitable for high-speed buffering of instrument data, and 256 MBytes of Triple-Modular-Redundant Flash, used to store up to eight separate software boot images [28].
The remaining 5 slots included SACI board, which was responsible for the solar panels and battery interface, and 2 PAPI and 2 SATORI boards, which are responsible for triggering the power for different actuators and sensors.All of the these cards communicated through the common bus provided by the IAU.
Due to the mission's low budget, the RAD750 and I/O boards have no backups. Instead, fault tolerance relied on software (memory scrubbing, fault detection) rather than hardware redundancy

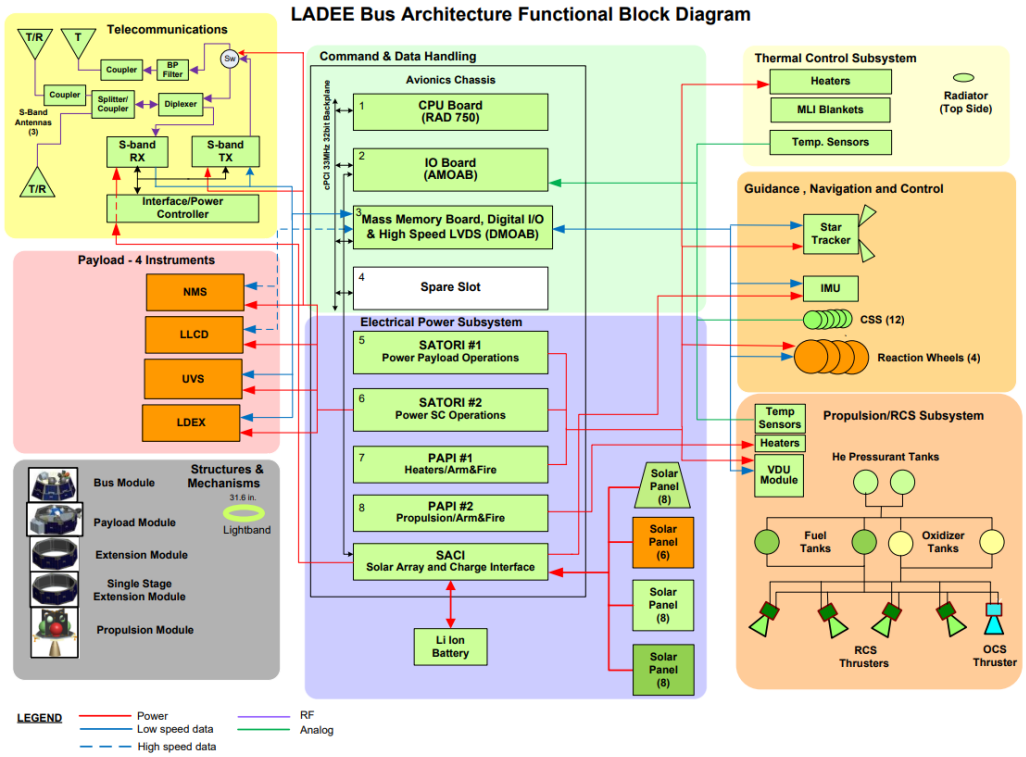


Figure 16: LADEE's bus architecture

8.1 OBDH Components

The core of LADEE's OBDH relied on a set of well-defined processing and control components, distributed between the central avionics system and the instrument subsystems.

- The primary flight processor was a radiation-hardened RAD750 microprocessor, operating at 132 MHz and capable of approximately 260 MIPS. It was supported by 128 MB of SDRAM for volatile memory operations and a 256 kB SU-ROM for non-volatile boot and fallback code storage [4]. The RAD750 hosted the main flight software, running the VxWorks real-time operating system, and was responsible for executing C&DH tasks, managing telemetry, and coordinating system-level functions across all spacecraft subsystems.
- NMS featured an independent command and data handling module based on a radiation-hardened 32-bit ColdFire microprocessor. This processor was paired with 2 MB of SRAM for flight software execution and temporary data buffering, 1 MB of EEPROM for persistent configuration and parameter storage, and a 32 kB programmable ROM for embedded firmware [22]. The NMS module autonomously handled sensor control and data packaging prior to uplink to the spacecraft OBDH.
- UVS utilized a Microchip RTAX2000S radiation-tolerant FPGA for instrument control. This antifuse-based device featured approximately two million system gates and supported up to eight I/O banks, enabling interface with various voltage domains and digital standards [40]. Within the FPGA, an embedded 8051 microcontroller core operated using firmware stored in an external 32k $\times$ 8 PROM. The FPGA was responsible for managing timing, triggering operations, and serial communication (via RS-422) with the spacecraft's avionics [22].
- LDEX employed an FPGA-based controller for managing its sensor operations. Although the specific model is not identified in the literature, it is assumed to be consistent with the RTAX2000S architecture used in UVS, given the shared design philosophy across instruments. Collected data were temporarily stored in SRAM prior to transmission to the central OBDH for downlink [22].

8.2 Data Bus

Internally, within the Integrated Avionics Unit (IAU), the CompactPCI (cPCI) backplane bus served as the high-speed communication backbone. This bus architecture enabled the RAD750 processor to interface with other cards installed in the chassis, including memory interface boards for the science instruments and power distribution units. The backplane operated on a 32-bit, 33 MHz link [22].

For external data communication, a UART-based serial protocol was used between the spacecraft and its subsystems. These serial lines were implemented over the RS-422 physical layer, which employs differential signaling to improve robustness against electromagnetic noise, which is an important advantage in the space environment. RS-422 also supports data rates up to 10 Mbps [61], which is significantly higher than LADEE's operational requirements as specified in Section 8.3.

For example, RS-422 served as the communication interface between the Command and Data Handling (C&DH) unit and the Micro µSTDN S-Band transponder [22]. Similarly, telemetry and science data from payload instruments such as LDEX, NMS, and UVS were transmitted to the central avionics system over RS-422 links [22].

8.3 Data Size on Board

For the data type and rate of the onboard parameters, several assumptions were made; For example, parameters like the pressure, temperature, voltages, and status of different components had values

that were obtained from a reference textbook since the actual values are not provided, taking into consideration their count.
The UVS provided raw data only, the frequency rate for this parameter was approximated to be the number of records (1038101) [59] divided by the instrument total operating hours (608 hours) [15]. The same was done for NMS, 5116 records [5] in 1008 hours [15]. as for NMS the data rate, the total value was 3.5 kbits/s [22], which was then divided between the raw data and housekeeping data based on the final total size of each [9]. For LDEX, the number of records for housekeeping (36) and raw data (121) [65] were very low compared to the total operating hours (1931 hours) [15]. The data rate was also found by dividing the total size of the raw data and the housekeeping data each by LDEX total operating hours [65].
The frequency of data transmission was assumed to be every 7 hours since LADEE does telemetry downlink daily for 2-4 lunar orbits [17], selecting an average of 3 and assuming that only the s/c is visible to earth for half the orbit, we are left with 7 hours spacing between each transmission.

8.4 Discussion

From the table below we can see that our estimates of the data rates combined with a 400% margin yield a total data volume to be stored of about 484 MB, which is more than the 128 MB of SDRAM of the OBC, but still less than the 1 GB of memory available in the DMOAB. Below can be seen how the majority of the data to be stored is generated by the scientific payload, which interface with DMOAB and explains the need for an increased memory buffer. Furthermore the high capacity of such buffer would allow the system to store the data even if downlink every 7 hours wasn't available.

9 Mission Overall Comment

9.1 Understand and critically comment

9.1.1 Spacecraft Architecture and Structural Philosophy

The Lunar Atmosphere and Dust Environment Explorer (LADEE) mission was a testament to NASA's pursuit of lower-cost, rapid-development spacecraft. Its structural philosophy was anchored by the Modular Common Spacecraft Bus (MCSB), a lightweight carbon composite framework developed at NASA Ames Research Center. This MCSB formed the primary load-bearing backbone of the spacecraft. Its core functions included withstanding the demanding dynamic environment imposed by the Minotaur V launch vehicle, maintaining the spacecraft's overall structural integrity, and providing a standardized platform for the integration of various mission-critical modules. The design prioritized ease of manufacturing and assembly, contributing to the mission's cost and schedule goals.

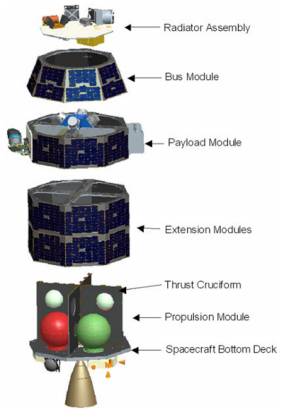


Figure 17: LADEE modules.

9.1.2 Structural Configuration of Spacecraft Modules

Building upon the primary MCSB structure, LADEE's configuration featured several distinct modules, largely classified as secondary structural elements, each dedicated to specific functions:

- The Radiator Module is housing the core avionics, the electrical power system, attitude sensors, and two of the science instruments (UVS and LDEX). It also incorporated essential thermal radiation surfaces.
- The Bus Module contained additional core spacecraft subsystems.
- The Payload Module was designed to accommodate the two most massive individual payloads: the NMS and the LLCD optical terminal. These were positioned within this module to optimize the spacecraft's center of mass.
- The Extension Modules and the integrated Propulsion Module provided the housing for the bi-propellant propulsion system, including tanks and thrusters.

Within these larger modules, a variety of internal brackets for securing electronics boxes, routing electrical harnesses, and general component mounting constituted the tertiary structural elements. These smaller structures were crucial for integration and local support but were not designed for significant global load-bearing. The geometry provides practical surface utility, as the flat sides support full coverage with body-mounted solar arrays, eliminating the need for deployable arrays—thereby simplifying the system and mitigating risks associated with vibrational modes typical of movable or flexible components.

9.1.3 Dominant Sizing Environments

The structural design of LADEE was driven by the launch phase environment. The Minotaur V launch vehicle, a stack of five solid rocket stages, subjected the spacecraft to significant axial and lateral accelerations, high-frequency random vibrations, and pyrotechnic shock events during stage

separations and fairing jettison. The MCSB and all module interfaces, particularly the Payload Attach Fitting, were engineered to safely manage and transmit these loads throughout the spacecraft structure without compromising integrity or alignment.

Following launch, on-orbit operational loads were a secondary but still important influence. These included the forces generated by firings of OCS thruster for OMMs and the smaller RCS thrusters for attitude adjustments and momentum desaturation. Additionally, the extreme thermal cycling experienced as LADEE transitioned between lunar day (direct solar illumination) and lunar night (deep cold) induced thermal stresses that the structure had to accommodate without damageful deformation. The End-of-Life (EOL) phase, which involved a controlled impact with the lunar surface, did not impose unique structural sizing requirements beyond those needed to survive the launch and the extended operational lifetime.

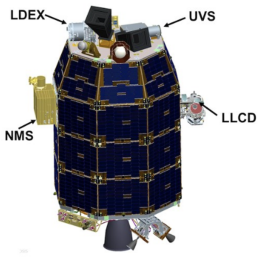


Figure 18: Instruments locations.

9.1.4 Mechanism Utilisation and Design Philosophy

LADEE's design philosophy emphasized minimizing mechanical complexity to enhance overall mission reliability and adhere to its low-cost mandate. This resulted in a limited use of mechanisms:

- Instrument Covers: Critical one-time deployment mechanisms were implemented for the science instruments. The NMS featured a pyrotechnically actuated break-off cap. Both the UVS and LDEX instruments had protective aperture doors deployed using TiNi pinpullers and spring-loaded hinges. The primary function of these covers was to shield sensitive optical surfaces and instrument internals from particulate and molecular contamination during ground operations, launch, and the initial on-orbit outgassing period.
- LLCD Pointing System: The LLCD payload was the most mechanically complex, incorporating an articulated optical modem mounted on a precision two-axis gimbal. This mechanism was essential for accurately pointing the laser beam towards Earth-based ground stations for the high-bandwidth optical communication experiment, operating independently of the main spacecraft's attitude.
- Internal Mechanisms: Standard internal mechanisms included the reaction wheels, electro-mechanical devices critical for 3-axis attitude control and stabilization, and numerous valves within the bi-propellant propulsion system for precise control of fuel and oxidizer flow to the thrusters.

9.1.5 Spacecraft Mass Breakdown

LADEE's mass budget was tightly controlled due to its classification as a small explorer mission. The total launch mass was approximately 383 kg, with a dry mass of about 248.2 kg. It carried roughly 134.8 kg of propellant (MMH fuel and MON-3 oxidizer) and Helium Pressurant. The payload, consisting of the science instruments and the LLCD, weighed around 49.6 kg.

9.1.6 Launcher Interface

LADEE was launched atop a Minotaur V vehicle, interfacing specifically with its Star 37 fifth stage. This connection was made via a Payload Attach Fitting (PAF), likely a standardized or semi-standardized design. The PAF served as the crucial structural link, transferring all launch-induced loads—axial and lateral accelerations, vibrations, and shocks—from the launch vehicle to the spacecraft. The LADEE structural design ensured that these loads were efficiently distributed from the PAF into the primary MCSB structure, typically through a robust ring or plate at the spacecraft's base, an area also housing the main propulsion elements. Strategic placement of high-mass components, like the fully fueled propellant tanks, near this interface was critical for minimizing bending moments and adhering to the launcher's strict center of mass envelope.

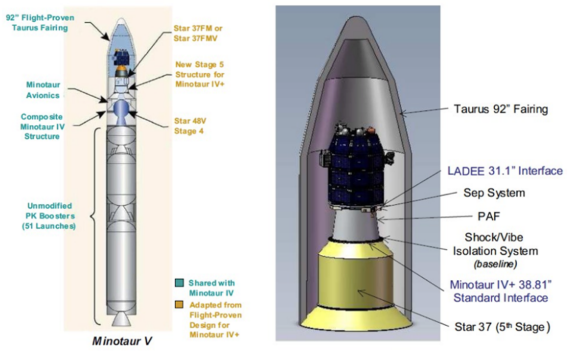


Figure 19: LADEE launch vehicle and configuration.

9.2 Design Justification and Analysis

9.2.1 Overall Vehicle Shape and Appendages

The LADEE spacecraft adopts a compact octagonal prism geometry, measuring 2.37 m in height and 1.85 m in maximum diameter. Its structure is composed of lightweight carbon composite panels, offering both mechanical strength and mass efficiency.

9.2.2 Shape Selection Rationale

The design of the MCSB is driven by requirements for modularity, manufacturability, and functional integration. Internally, the octagonal prism shape yields a total volume estimated as [4]:

$$V = 2(1 + \sqrt{2}) \cdot a^2 \cdot h = 5.73\,\mathrm{m}^3 \tag{42}$$

where $a = 0.7079\,\mathrm{m}$ is the apothem and $h = 2.37\,\mathrm{m}$ the total height of the structure. This configuration ensures sufficient internal space for the accommodation of all subsystems and payload components without excessive bulk or structural overdesign.

9.2.3 Structural Payload Allocation

The LADEE structure was optimized to carry science payloads in two modules:

- **Payload Module:** Accommodates the LLCD (30.7 kg) and NMS (11.8 kg) instruments [22], with a structural payload limit of 50 kg [67]. In this section the margin of safety adopted is 17%;
- **Bus Module:** Hosts lighter payload instruments, such as UVS and LDEX (3.6 kg each). Based on configuration data, a 10 kg limit for the payload is assumed as the lower bound for safe operation considering the simpler and lighter Lunar Lander configuration of the MCSB [67]. Also in this section the margin of safety adopted is 38%.

These margins reflect an efficient yet non-excessive structural design, tuned to minimize mass while maintaining functional integrity.

[4] This refers only to the prism body and excludes the truncated pyramid top; internal structures also limit the usable volume for individual components.

9.3 Launcher Integration: Interfaces and Fairing Constraints

9.3.1 Mechanical and Electrical Interfaces

The LADEE spacecraft was mounted to the Minotaur V launch vehicle via a standard mechanical interface, consisting of either a 944 mm or 986 mm bolt circle, using a Marmon-style clamp band separation system. This system utilized dual semi-circular stainless steel bands with aluminum separation rings. Redundant bolt cutters initiated separation, and stored energy springs provided a separation impulse of up to 37.6 J. The interface hardware attached to the spacecraft post-separation weighed approximately 5.7 kg for the 986 mm interface or 4.6 kg for the 944 mm variant.

9.3.2 Fairing Configuration & Structural and Environmental Constraints

LADEE was housed inside the 92" (2.34 m) diameter standard Minotaur fairing. The payload envelope allowed for a maximum diameter of 2.055 m and a usable vertical height of 4.834 m from the payload separation plane to the fairing tip. The fairing was constructed from graphite/epoxy composite face sheets with an aluminum honeycomb core, providing both structural integrity and low outgassing. Internal surfaces were lined with acoustic blankets, and conditioned air maintained temperatures between $13\,^{\circ}\text{C} - 29\,^{\circ}\text{C}$ during ground operations. A single 457 mm × 610 mm access door was standard, with options for additional doors available upon request. The structural design of LADEE was subject to stringent constraints dictated by the Minotaur V environment:

Constraint	Requirement
Acceleration Loads	Up to +5 g axial, 3.5 g lateral during ascent
Random Vibration	12.6 RMS g from 20 Hz to 2000 Hz
Acoustic Environment	Max 138 dB OASPL inside fairing
Shock Environment	Up to 1000 g at separation (interface-dependent)
Center of Gravity (C.G.)	Within 6.35 mm of nominal in all axes
Mass Property Accuracy	±0.5% for mass, ±5% for inertia matrix
Modal Requirements	First bending mode > 25 Hz; axial modes outside 35 Hz to 45 Hz or > 75 Hz

Table 29: launch environment constraint.

9.3.3 External Payload and Sensor Placement

All payloads were externally body-mounted along with the star tracker heads and antennas. Most instruments were positioned on the top radiator of the MCSB to avoid thruster plume interference. The only instrument near the propulsion system was a low-gain omni-directional antenna, ensuring full coverage in any attitude. With pointing managed by the ADCS, each sensor retained its own FOV. The fixed solar panels posed no obstruction to instrument fields of view.

9.3.4 Internal Equipment Layout and Mass Balancing

Internally mounted instruments and actuators were arranged to keep the center of mass aligned with the main thruster axis. LADEE's near-axial symmetry simplified attitude control. High-power and heat-generating components, such as avionics and power electronics, were mounted near radiative surfaces, while thermally sensitive units were thermally isolated. Carbon composite panels with embedded foil provided lightweight structural support, grounding paths, and charge dissipation. This zoned design supported efficient thermal management and protection against deep charging in the lunar environment.

10 Mass budget tables

10.1 Mass budget from sizing

Subsystem	Component	Mass[kg]	Margin [%]	Mass + Margin [kg]	Reference
PS	Pressurizer Tanks	1.1096	10	1.2329	H2
	Fuel Tanks	0.5288	10	0.5875	H2
	Oxidizer Tanks	0.5288	10	0.5875	H2
Propellant	Fuel	51.9	-	51.9	H2
	Oxidizer	84.9	-	84.9	H2
	Pressurizer (Regulated)	0.4046	20	0.4855	H2
	Pressurizer (Blowdown)	0.1366	20	0.1639	H2
TMTC	-	-	-	-	-
AOCS	-	-	-	-	-
TCS	MLI	-	-	-	-
	Coatings	Negl.	-	Negl.	-
	Radiators	27.98	20	34.98	H5
	Conductive Connections	-	-	-	-
	Isolative Connections	-	-	-	-
	Heaters	2	20	2.4	H5
EPS	Battery	9.097	-	9.097	H6
	SA	27.53	-	27.53	H6
	Cables	-	-	-	-
OBDH	-	-	-	-	-
Payload	-	-	-	-	-
Other	-	-	-	-	-

Table 30: Mass Budget per component from sizing.

The empty rows in the table 30 represent subsystems or components whose mass values were not obtained directly from sizing. The following table 31 groups all the subsystems' mass obtained from sizing completed with literature values.

Component	Mass w/o Margin [kg]	Margin [%]	Mass with Margin [kg]
Propulsion Subsystem (dry)	13.4	10	14.1
TMTC Subsystem	4.2	10	4.6
AOCS Subsystem	15.7	0	15.7
TCS Subsystem	30.7	20	36.8
EPS Subsystem	37.1	5	38.9
Payload	49.6	0	49.6
Other Dry Mass Components	90.0	20	108.0
Total Dry Mass	240.7	—	268.7
Propellant Mass	136.8	0	136.8
Wet Mass	377.5	—	405.5
Launchable Mass	377.5	5	396

Table 31: Mass Budget Summary from system sizing (RE).

10.2 Mass budget from literature

Subsystem	Component	Mass[kg]	Margin [%]	Mass + Margin [kg]	Reference
PS	Feeding System	4.22	10	4.642	[62][57]
	Primary Thruster	5.2	0	5.2	[42]
Propellant	Fuel	51.2	-	51.2	[22]
	Oxidizer	82.1	-	82.1	[22]
TMTC	S-band Transponder	2.1	0	2.1	[63]
	MGA	1.0	10	1.1	Assumed
	2×LGA	2×0.6	10	2×0.66	Assumed
AOCS	12×Sun Sensor	12×0.024	0	12×0.024	[7]
	2×Star Tarcker	2×3.9	0	2×3.9	[60]
	4×IMU	RW integrated	0	RW integrated	[41]
	4×RWs	4×1.44	0	4×1.44	[41]
	4×Secondary Thrusters	4×0.454	0	4×0.454	[42]
TCS	MLI	6	20	7.2	[45]
	Conductive Connections	0.7	20	0.84	[32]
	Isolative Connections	0.7	20	0.84	[32]
EPS	Battery	6.5	-	6.5	[22]
	Cables	4.5	20	5.4	[38]
OBDH	Central OBC	6.26	0	6.26	[36]
	Data Bus and Interfaces	2	10	2.1	Assumed
Payload	LLCD	30.7	0	30.7	[22]
	NMS	11.8	0	11.8	[22]
	UVS	3.6	0	3.6	[22]
	LDEX	3.6	0	3.6	[22]
Other	Bus Structure	90	20	108	[46]
	Misc.	15	20	18	Assumed

Table 32: Mass Budget per component from literature.

Table 32 represents only subsystems or components whose mass values were obtained directly from literature. The following table 33 groups all the subsystems' total mass obtained using literature values.

Component	Mass [kg]	Reference
Propulsion System (dry)	25	[42, 22]
TMTC Subsystem	4.5	[63]
AOCS Subsystem	15.7	[42, 60, 7]
TCS Subsystem	9.0	[45, 32]
EPS Subsystem	18.0	[22, 38]
Payload	49.6	[22]
Other Dry Mass Components	105.0	[46]
Total Dry Mass	248.2	[22]
Propellant Mass	134.8	[22]
Wet Mass	383.0	[22]
Launchable Mass	383.0	[22]

Table 33: Mass Budget Summary from literature.

10.3 Mass Budget Considerations and Assumptions

The tables provided report the mass breakdown of the spacecraft distinguishing between values obtained from literature and those derived from system sizing. A number of important assumptions and clarifications are necessary to properly interpret the values presented, especially where discrepancies or missing data appear.

10.3.1 Propulsuive Subsystem

No direct mass margins were applied to the tanks. Instead, a margin of 10% was introduced on the design volume during the sizing process. The final tank masses were then derived from this increased volume, effectively embedding the margin within the derived mass. Therefore, tank masses are considered as already accounting for volumetric and pressure uncertainties. The pressurizing system transitions between: pressure-regulated and blowdown. Both systems were dimensioned properly, and the reported masses in the sizing table reflect the hardware required for each configuration. The reported mass values include margins of 20%, accounting for potential variability in the design. The feeding system mass in the first table is sourced from literature and includes valves and pipelines. These components are based on off-the-shelf solutions, whose mass has been preserved without added margin. Both fuel and oxidizer quantities are given as nominal required propellant mass, without added margins, as these are considered mission-driven quantities. However, as mentioned above, a 10% volumetric margin was applied to the tanks, ensuring that real storage requirements are met with operational margin. Thus, the tanks accommodate slightly more than the bare minimum mass of propellants.

10.3.2 TMTC Subsystem

The MGA and LGA were mission-specific designs. Since no off-the-shelf mass data was available, assumed values were used, based on previous missions and engineering judgment. In particular, the mass of two LGAs was taken as 2×0.6 kg, with an applied 10% margin.

10.3.3 Electrical Power Subsystem

The cabling is estimated based on wire length and type. According to the cited reference, the system likely uses between 100–300 meters of AWG20 wire, suitable for currents up to 6.5 A. Given a specific mass of 0.02 kg/m, the total cable mass can be expected in the 2–6 kg range, with 4.5 kg being a reasonable conservative estimate. A 20% margin was applied to account for routing complexity and harness interfaces.

10.3.4 Thermal Control Subsystem

The radiator and heater masses included in the sizing table result directly from the area and power requirements derived in a dedicated paragraph. Specifically, radiator sizing was driven by heat rejection needs, and their mass includes structural support and surface coatings. Similarly, heater mass includes integrated electronics and control elements. A 20% margin was applied to both components, as per standard practice.

10.3.5 Miscellaneous and Bus Structure

The "Miscellaneous" mass is intended to capture non-modelled or minor components not explicitly represented in the breakdown. Based on spacecraft design heritage and preliminary estimates, it was fixed to represent approximately 7% of the total dry mass. This value is consistent with standard spacecraft-level budgeting practices.

11 Appendix

DATA BUDGET BREAKDOWN					
Parameter name	**n.**	**bit size**	**variable type**	**frequency [Hz]**	**data rate [bps]**
EPS					
Battery voltage	1	16	int_16	1	16
Battery current	1	16	int_16	1	16
Battery temperature	1	16	int_16	1	16
BUS voltage	1	16	int_16	1	16
OBDH subsystem voltage	1	16	int_16	1	16
Solar panel voltage	12	16	int_16	1	192
Solar panel current	12	16	int_16	1	192
Solar array switches	12	1	bool	0	0
NMS, UVS, LLCD, LDEX status	4	1	bool	0	0
IMU, Star Tracker status	2	1	bool	0	0
OVD Module status	1	1	bool	0	0
Propulsion Subsystem					
Fuel tank pressure	2	16	int_16	1	32
Oxidiser tank pressure	2	16	int_16	1	32
Pressurant tank pressure	2	16	int_16	1	32
Fuel tank temperature	2	16	int_16	1	32
Oxidiser tank temperature	2	16	int_16	1	32
Pressurant tank temperature	2	16	int_16	1	32
Fuel mass	2	32	float_32	1	64
Oxidiser mass	2	32	float_32	1	64
Fuel Valve	1	1	bool	0	0
Oxidiser Valve	1	1	bool	0	0
Main thruster status	1		bool	0	0
Engine Temperature	1	16	int_16	1	16
AOCS					
Sun sensor (angle)	12	16	int_16	10	1920
Star tracker (quaternion)	4	32	float_32	10	1280
Single axis gyro (angular speed)	4	32	float_32	100	12800
Reaction Wheel (Torque)	3	32	float_32	10	960
Thrusters (Torque)	3	32	float_32	10	960
Thrusters Temperature	4	16	int_16	1	64
RW Temperature	4	16	int_16	1	64
Gyroscopes status	4	1	bool	0	0
IMU temperature	1	16	int_16	1	16
AOCS mode	1	8	int_8	0	0
Star tracker temperature	1	16	int_16	1	16
Payload					
UVS raw data	1	----------	----------	0.5	9000
NMS raw data	1	----------	----------	0.00035	3120
NMS housekeeping data	1	----------	----------	0.00035	380
LDEX raw data	1	----------	----------	----------	4313.251626
LDEX Housekeeping	1	----------	----------	----------	415.4439266
NMS, UVS, LLCD, LDEX temperature	4	16	int_16	1	64
OBDH					
CPU voltage	1	16	int_16	1	16
CPU temperature	1	16	int_16	1	16
MOAB FPGA voltage	1	16	int_16	1	16
MOAB FPGA Temperature	1	16	int_16	1	16
TCS					
Avionics box temperature	1	16	int_16	1	16
external temperature sensors	12	16	int_16	1	192
NMS, UVS, LLCD, LDEX Heaters	4	1	bool	0	0
Battery heater	1	1	bool	0	0
Avionics box heater	1	1	bool	0	0
Tanks heaters	4	1	bool	0	0
IMU and Star Tracker heaters	2	1	bool	0	0
Radio heater	1	1	bool	0	0
TMTC					
Radio mode	1	8	int_8	0	0
upload	1	----------	----------	----------	2000
Radio Temperature	1	8	int_8	1	8

Total Variable size	645	bit		
Total data rate	38452.7	bit/s		
Storage time	7	h	*assuming transmission 3 times a day	
Storage required	9.69E+08	bit	121.126	Mbyte
with margin	400.00%	%	484.504	Mbyte

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